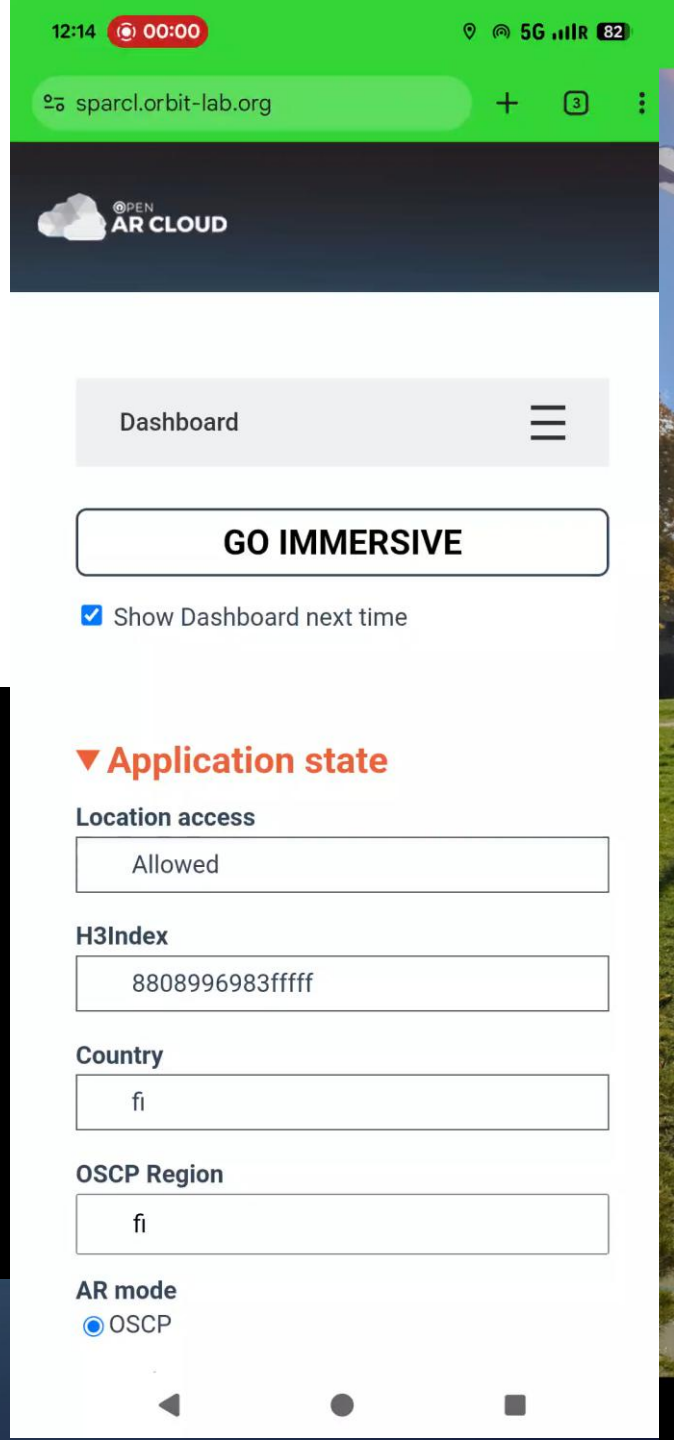
Demo at the campus

https://sparcl.orbit-lab.org

Try it!

Video: https://youtube.com/shorts/P_xfyFADVQU







UBICOMP ISWC2025

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Helsinki, Finland

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OARC WebXR Client

Dr. Gábor Sörös

Open AR Cloud / Nokia Bell Labs

OPEN AR CLOUD

Vision: AR cloud

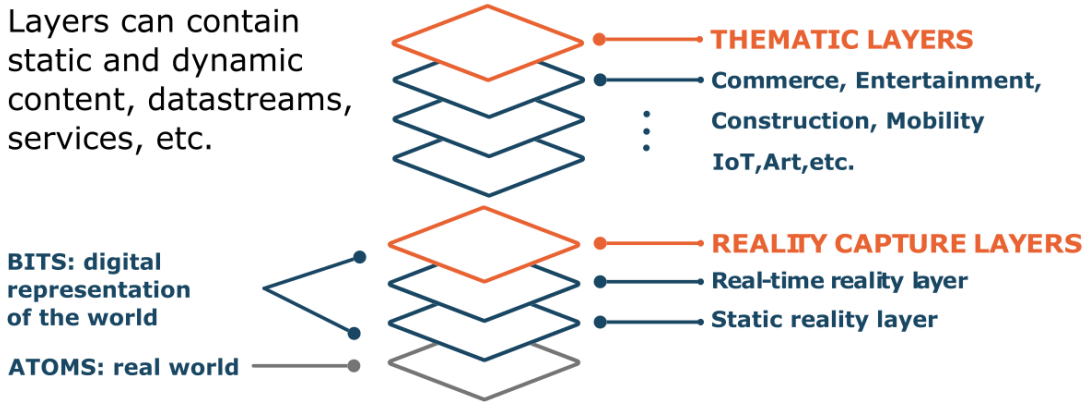
Augment the world with persistent digital objects

- Objects anchored to geo-locations
- Global 6DoF localization of client devices
- Collaborative mapping
- Geospatial query of contents
- Collaborative authoring: anyone can create
- Open: anyone can browse contents
- Real-time and interactive presentation

Open Web → Open Spatial Web?

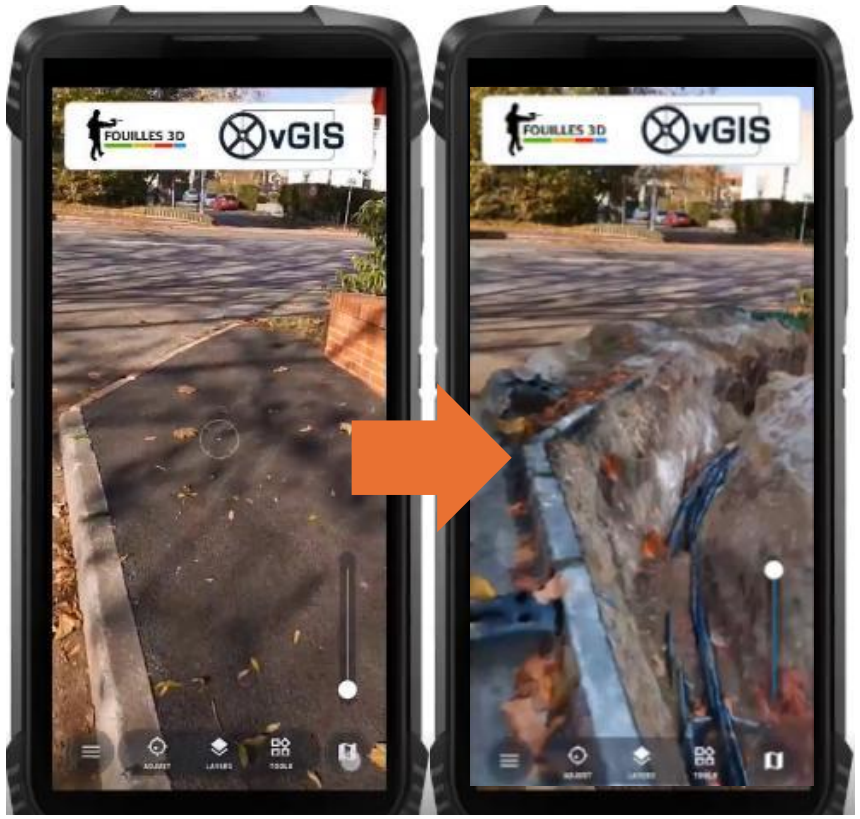


Source: https://www.rypplzz.com/images/westfield-05.jpg



Examples: real-time information & situated visualization

Location-based information on the go



vGIS Fouilles3D: GIS visualization



Immersal: train schedule



Snap: restaurant data in London

Why WebXR?

WebXR

+ runs in browser, no need to install anything
+ triggering AR experiences becomes very easy, seamless activation
+ universally browsing various types of digital contents
+ (ideally) wide range of devices supported

- but: at experimental stage, very limited browser support
- currently **Android Chrome** full functionality, Edge partial functionality, Safari ??
- starting the camera requires explicit button press, and camera-access feature is mandatory



SPARCL = spatial AR cloud client

`https://github.com/OpenArCloud/sparcl`

Digital contents attached to GeoPose

- primitives from textual descriptions
- GLTF/GLB models (texture OK, but no complex shaders)
- icons
- OGC Point of Interest
- iframe: complex scene loaded via URL
- OGC GeoPose streams → Metaverse Standards Forum



Try it yourself!

https://sparcl.orbit-lab.org

your own content here!

1. Enable WebXR in Android Chrome
`chrome://flags`
Search for ‘WebXR Incubations’, set to Enabled

2. spARcl is deployed at:
`https://sparcl.orbit-lab.org`
(use HTTPS)

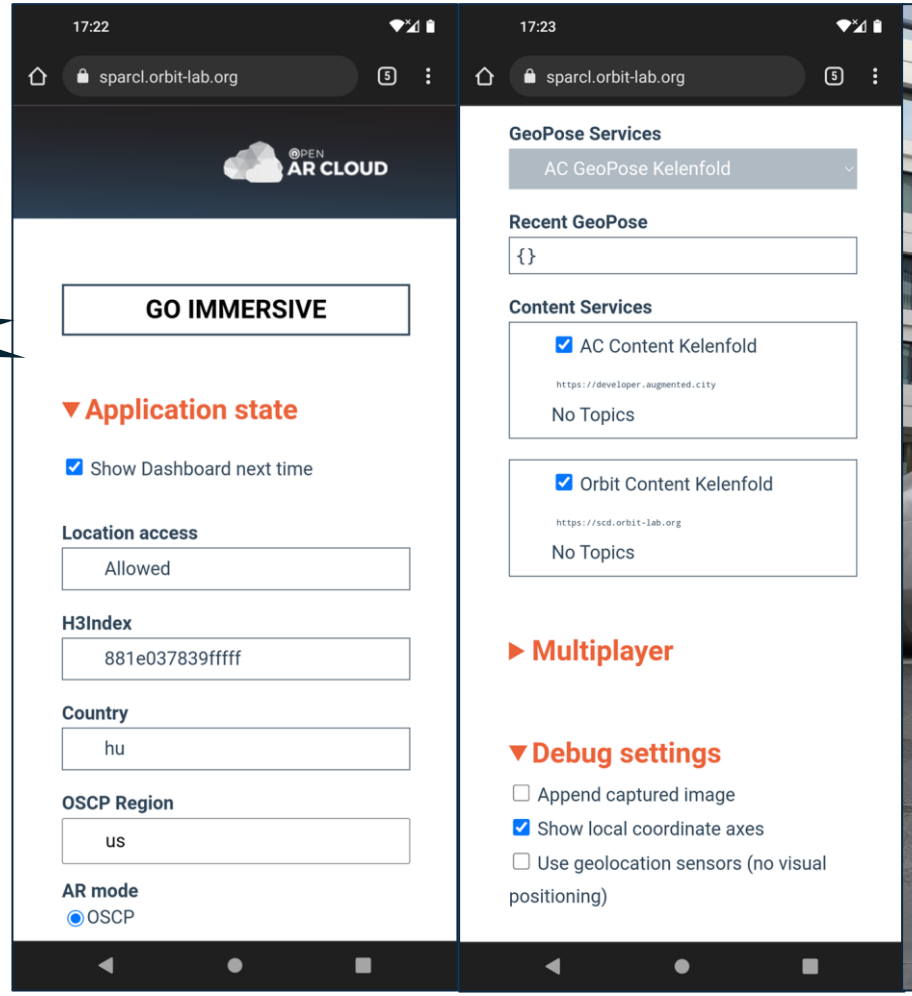
Try it!

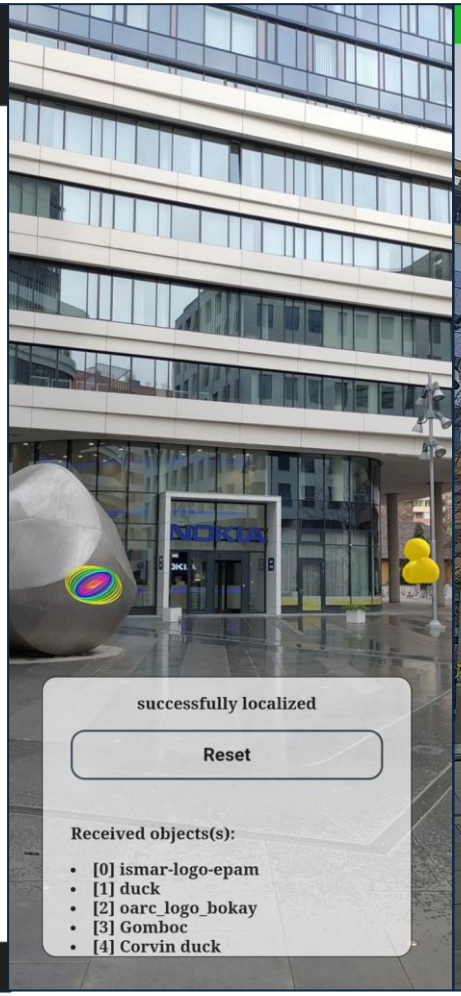
3. Enable Permissions
by clicking on the lock icon in the URL field (the app will also ask for them): Camera, Location

4. Swipe through the intro
the camera starts automatically. Press back to come to the Dashboard

5. Select services
review the gathered information. Select AC Geopose, Orbit content, XRM content, etc.

6. Go immersive
Look around, Localize, Enjoy







Setting up the code 1/2

Prerequisites:
git, npm (Node.js), code editor (ex. Visual Studio Code)

Code and dependencies:

```
cd ~/dev/OpenARCloud
git clone https://github.com/OpenArCloud/sparcl
cd sparcl
npm install
```

Enable local HTTPS server:
HTTPS is a must because we deal with camera and location access

Set Service Discovery root URL in .env file:
`VITE_SSD_ROOT_URL= <...>`

vite.config.js

```
plugins: [
        svelte(),
        glsl(),
        basicSsl(),
        typescript(),
    ],
```

Setting up the code 2/2

Enable WebXR in Android Chrome

`chrome://flags`

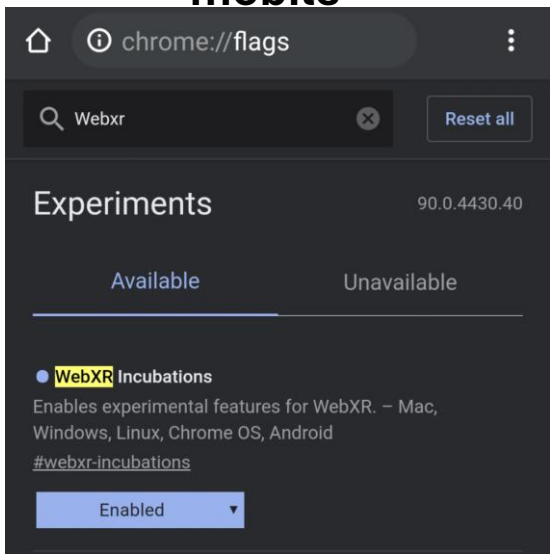
Search for ‘WebXR Incubations’, set to Enabled

Chrome remote debugging

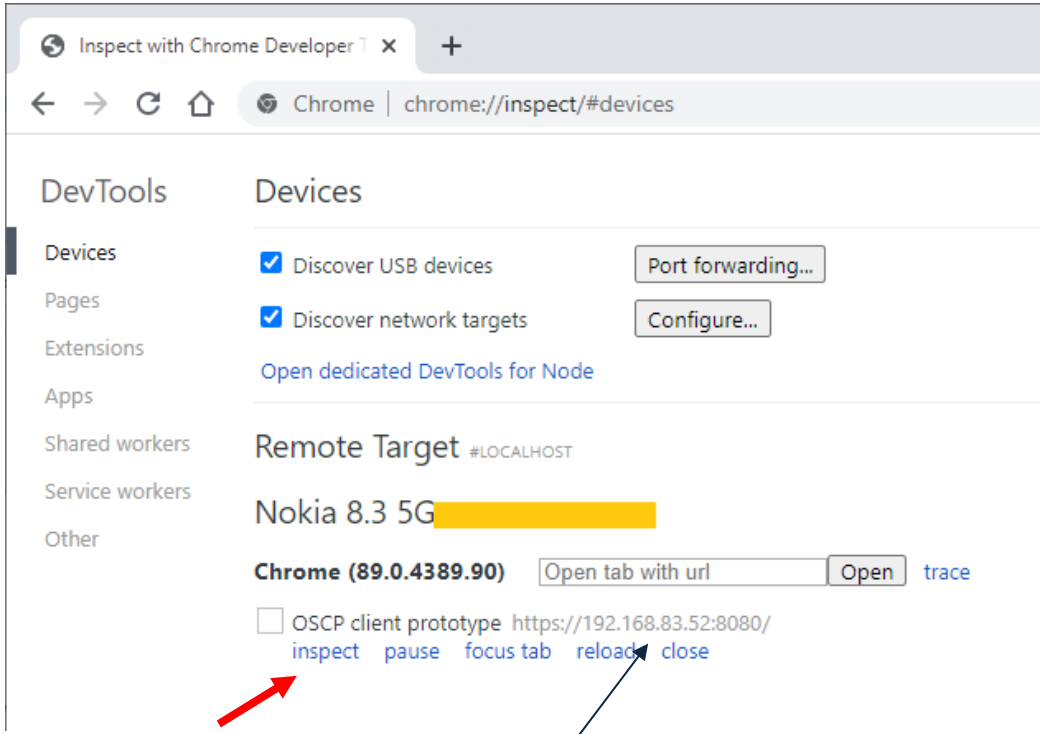
chrome://inspect/#devices

Attach phone via USB cable or enable Wireless debugging in the Developer options

mobile



desktop



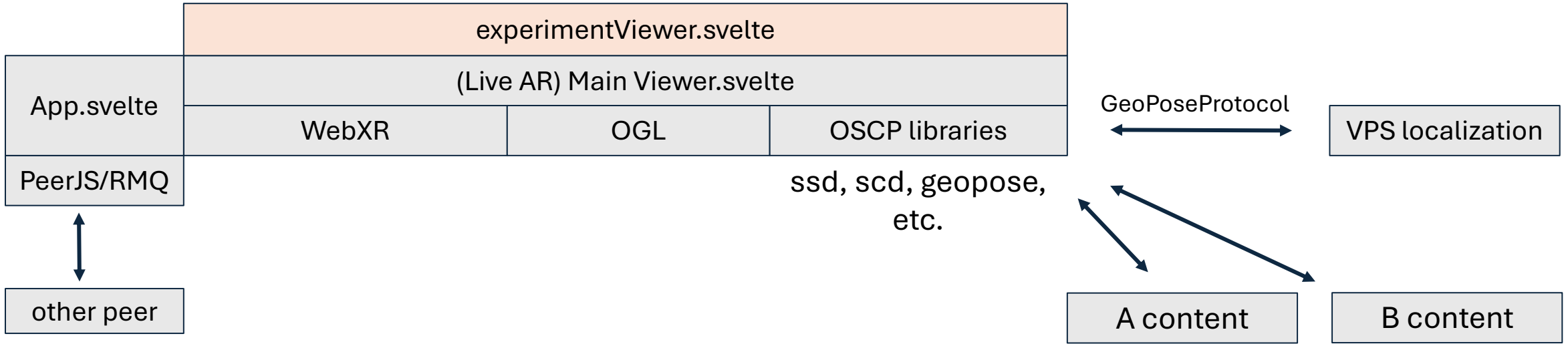
Finally, start the Web server:

```
cd ~/dev/OpenARCloud/sparcl
npm run start
```

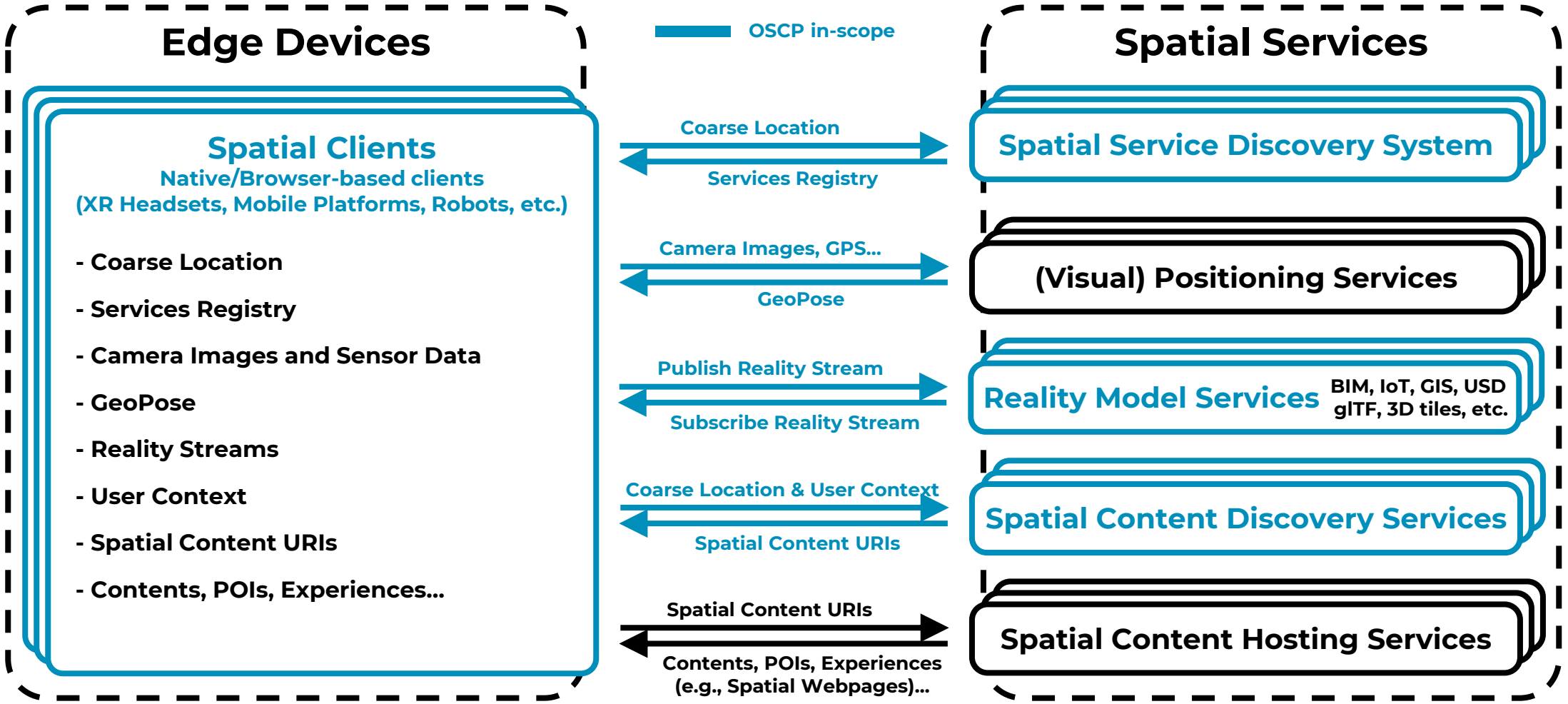
note the URL in the output (make sure your phone and server are on the same network)

Application architecture

Svelte: JavaScript frontend framework for building user interfaces with many advantages compared to Vue and React. Svelte code compiles to JavaScript DOM manipulations, avoiding the need to bundle a framework to the client, and allowing for simpler application development syntax.
Vite: fast build tool with nearly immediate reload or changed JavaScript modules without losing state
OGL: lightweight WebGL 3D engine
PeerJS or RabbitMQ: for multiplayer, will be covered later today



OSCP 3.0: SEQUENCE DIAGRAM



Authentication

Authentication via Auth0

- Required if we want to add contents in the future (but can also be hosted with auth disabled)
- Username automatically extracted

VITE_NOAUTH=true in env file disables authentication

We use Auth0.com for wrapping several OAuth2.0 providers. For example Google.

If you want to use Auth0, you can either create your own tenant and app, or ask us for our settings.

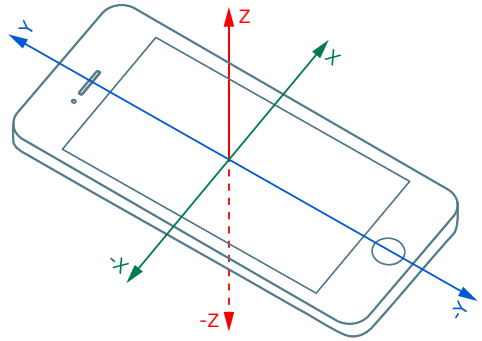
Localization with a supported VPS

You or we need to create the map of your environment first. This means capturing many photos and waiting a few hours.

- OpenVPS: StrayScanner https://youtube.com/shorts/oh5gkDY84FE
- AugmentedCity: AC Scanner https://youtube.com/shorts/U9Pmh6FyzO0
- (Immersal: Immersal Mapper) https://youtube.com/shorts/MqdGf44xn3o

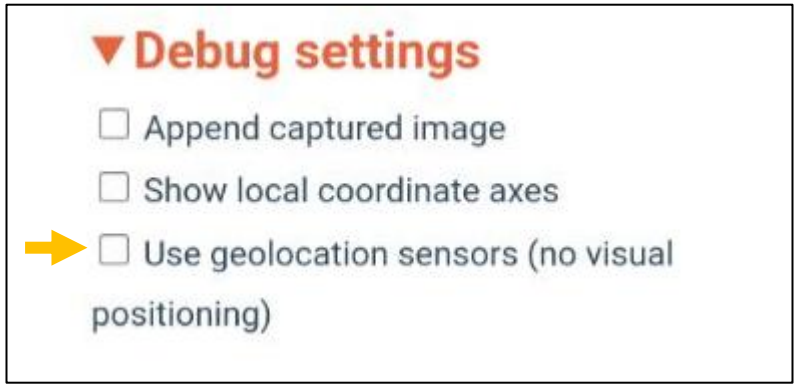
Please contact us for registering the selected VPS provider as a "GeoPose service" at your location to be listed in our spatial service discovery (SSD). Find your intended AR location on Google Maps, right click, and copy the GPS coordinates. We will enable the selected GeoPose service around that location.

Fallback: sensor-based localization



`GeolocationSensor` (fused GPS and RF signal strength)

`AbsoluteOrientationSensor` (fused acc, gyro, magnetometer)



Pros:

- works out of the box
- no mapping needed, no photos taken

Cons:

- inaccurate *(localize repeatedly and observe the deviations)*
- coordinate system may not be consistent across devices *(not suitable for multiplayer)*
- coordinate axes depend on screen orientations *(currently works only in landscape mode)* → *force landscape UI*

Camera access in WebXR

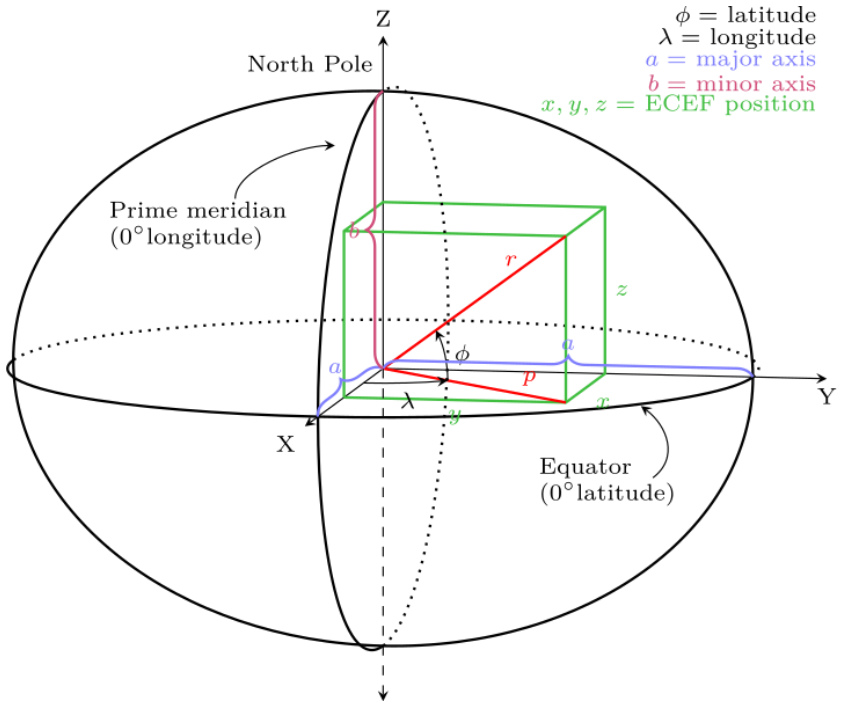
No access to images from the browser by default.

An extension allows to query the ID of the camera texture used for rendering the background. We can tap on that texture in GPU memory.

sparcl/src/core/cameraCapture.js

```
function createImageFromTexture(gl, texture, width, height)
```

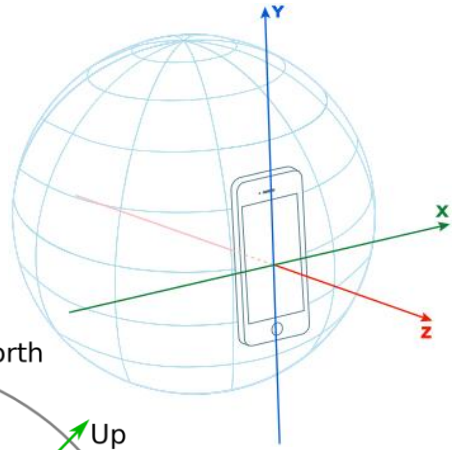
Coordinate systems

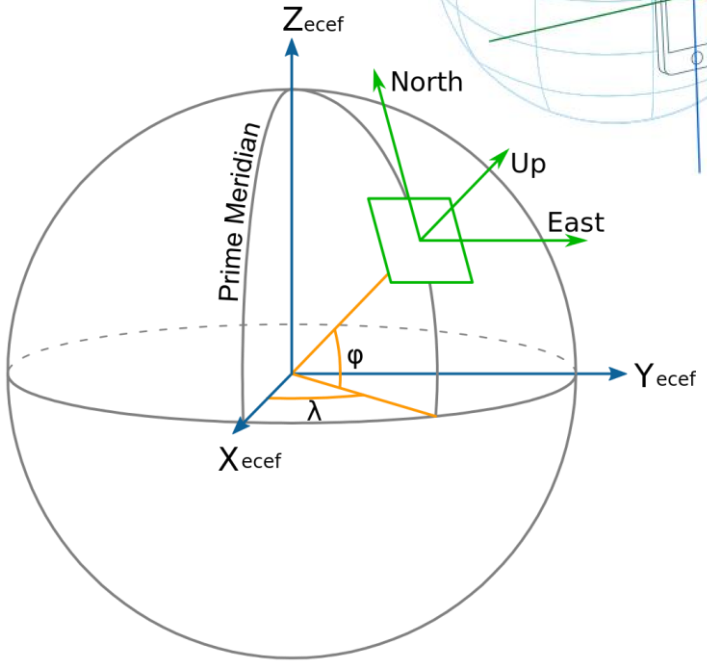


ECEF: Earth-Centered, Earth-fixed

used in GeoPose ecef field

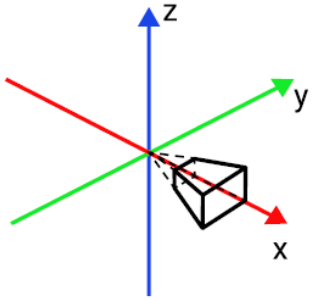
W3C OrientationSensor



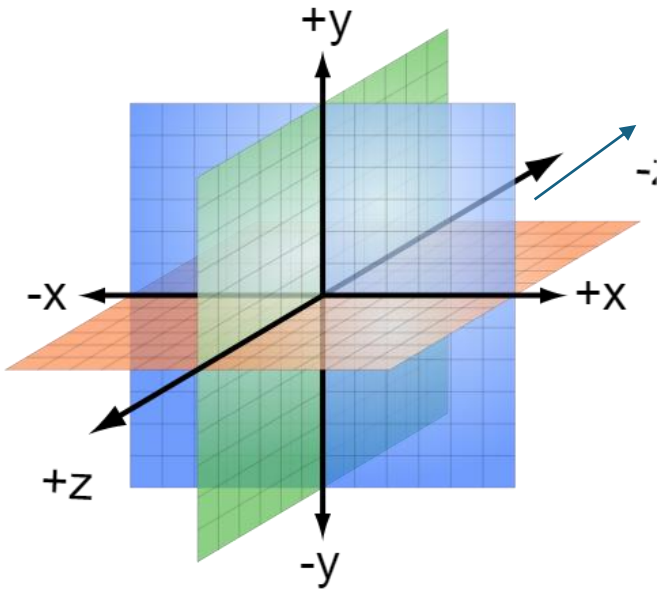


LTP: Local Tangent Plane approximation
ENU: East-North-Up (Z up)

used in GeoPose location field



GeoPose ENU
camera looks towards E/X



WebXR local coordinate system
right handed, Y up

used in GeoPose object orientation field

WebXR camera looks towards -Z

Querying *services* and *contents*

The services for an area need to be registered in a Spatial Service Discovery server

We run a sandbox SSD service at https://ssd.orbit-lab.org

Sparcl queries this server for available services around the user's coarse location

The content entries should be registered in a Spatial Content Discovery (SCD) server that covers the area.

We also run a sandbox SCD service at https://scd.orbit-lab.org

Sparcl queries this server for available contents around the user's coarse location

Sparcl (currently) is only for querying content, and it cannot add new contents to SCD!

We also provide a service/content administration GUI at https://oscp-admin.orbit-lab.org

H3 index

Hierarchical location index developed by Uber
Multilevel hexagonal grid, down to ~1 m^2 resolution
Easy search in all directions
Neighbor cells have the same distance

Allows search without exposing the user's GPS location

OSCP uses H3 indices in all queries
(next version will support general GeoJSON instead)

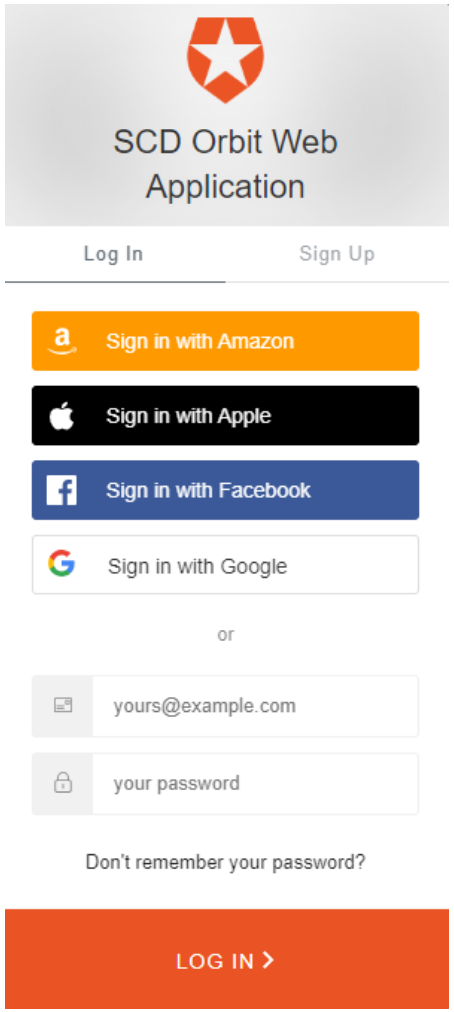


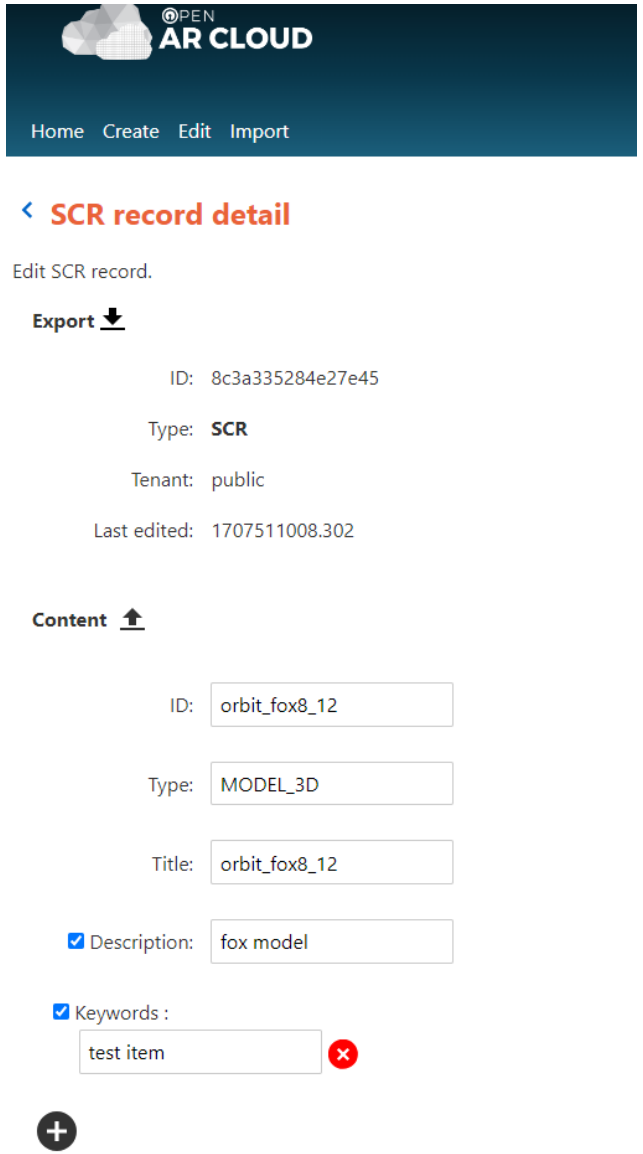
https://github.com/clupasq/h3-viewer

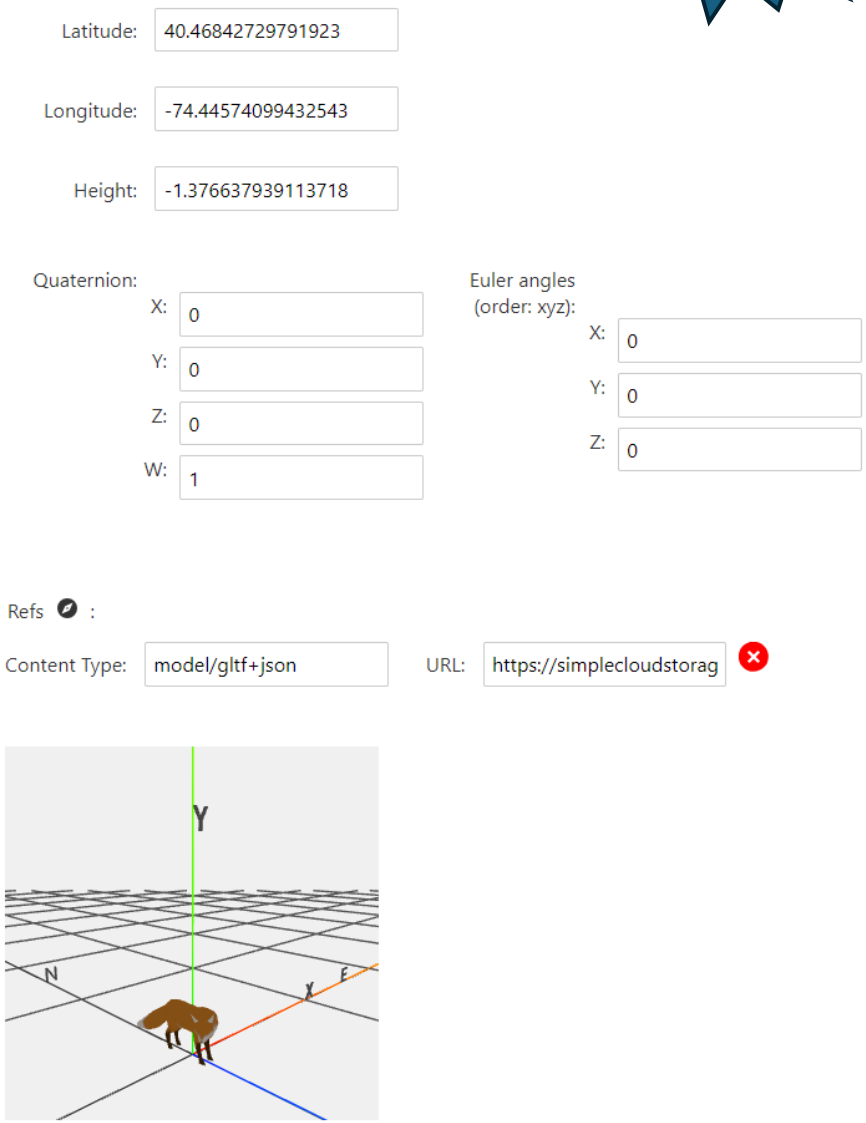
oscp-admin

https://oscp-admin.orbit-lab.org

Try it!



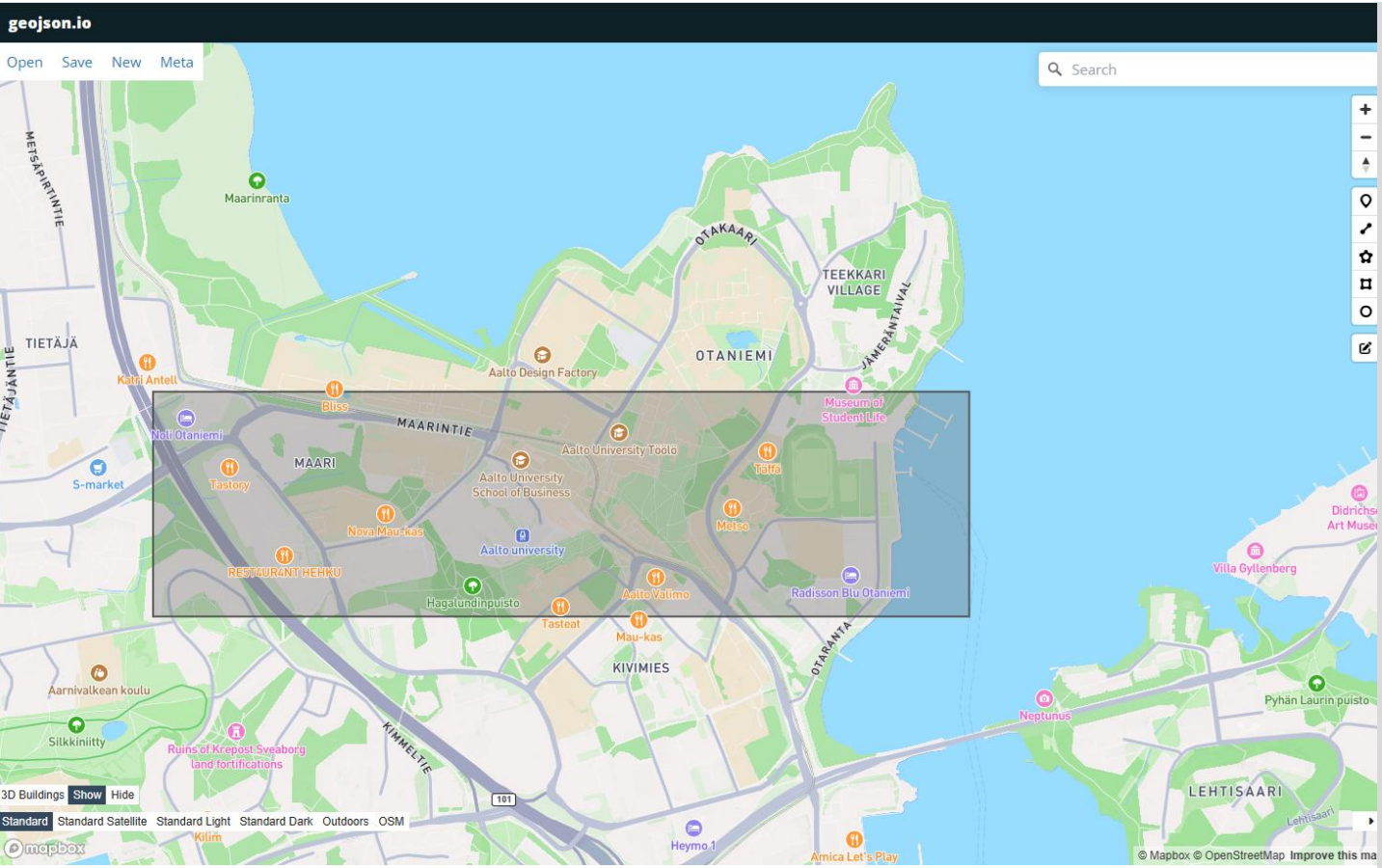




Test services at Aalto University

Coverage area as GeoJSON, Query area as H3 index

H3 index: 8808996983fffff

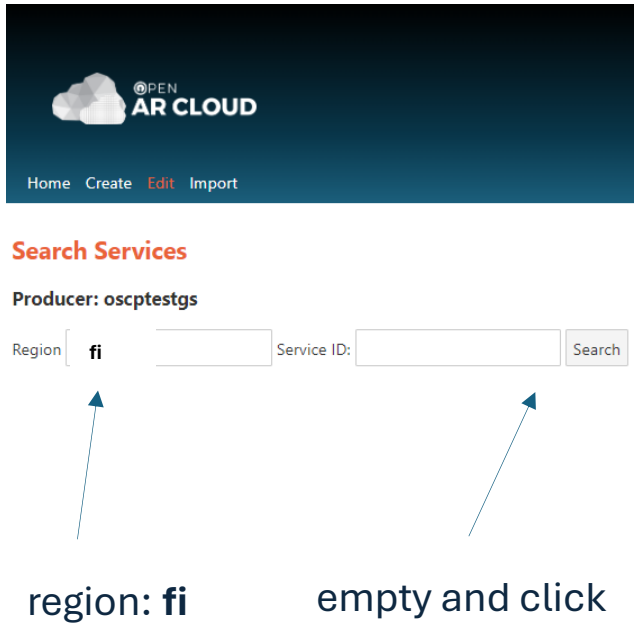


https://geojson.io/


```
{
  "type": "FeatureCollection",
  "features": [
    {
      "type": "Feature",
      "properties": {},
      "geometry": {
        "type": "Polygon",
        "coordinates": [
          [
            [
              24.80917438396444,
              60.187242412084935
            ],
            [
              24.80917438396444,
              60.1828892548095
            ],
            [
              24.84100095181705,
              60.1828892548095
            ],
            [
              24.84100095181705,
              60.187242412084935
            ],
            [
              24.80917438396444,
              60.187242412084935
            ]
          ]
        ]
      }
    }
  ]
}
```

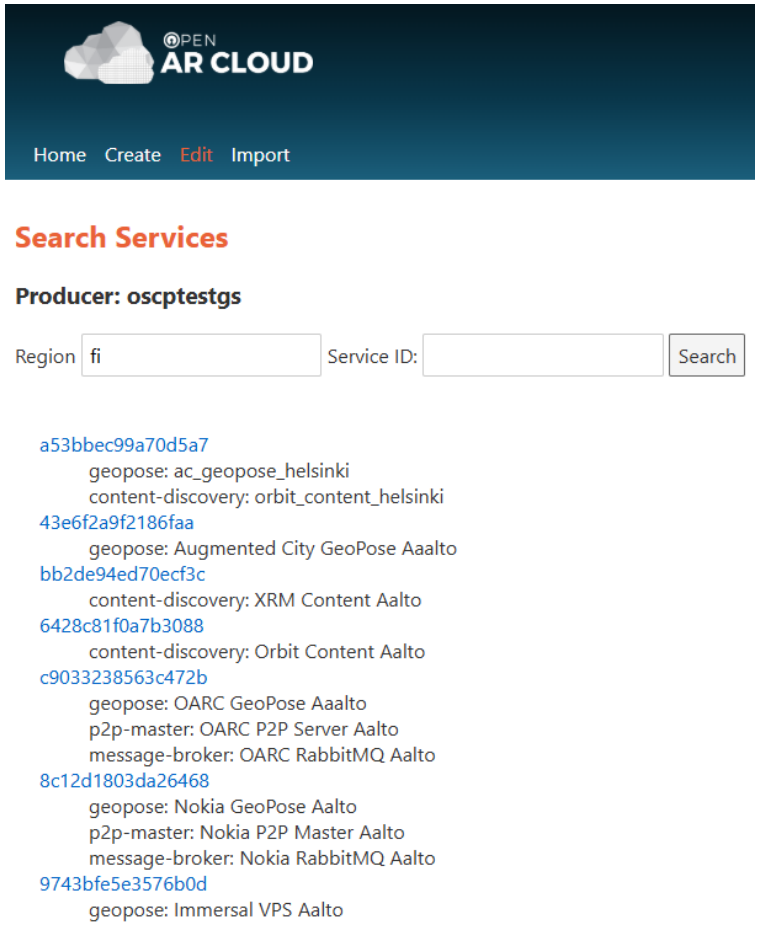
Test services at Aalto University

Create SSR



region: **fi**

empty and click



https://oscp-admin.orbit-lab.org/ssd

Test services at Aalto University

Search and validate

Now we can try to query these services:

a) by clicking Edit in `https://oscp-admin.orbit-lab.org/ssd`
setting **fi** as region, and clicking search

b) via Swagger GUI: `https://ssd.orbit-lab.org/swagger/`
setting **fi** as region and 8808996983fffff as H3 index

c) directly from SSD: `https://ssd.orbit-lab.org/fi/ssrs?h3Index=8808996983fffff`
note that client apps assemble this URL and use SSD this way

Test content at Aalto University

Create SCR

only
history
(for Database)

Create Spatial Content Record

Enter data for new SCR record.

Export

Type: **SCR**

Topic: history

SCR record detail

Edit SCR record.

Export

ID: 945790527ffa2215

Type: **SCR**

Tenant: public

Last edited: 1760173138.131

your ID for this object

Content

ID: aalto_dragon

only
MODEL_3D
(for Sparcl)

Type: MODEL_3D

Title: aalto_dragon

Description: A dragon at Aalto

Keywords :

Placekey:

https://oscp-admin.orbit-lab.org/scd

GeoPose

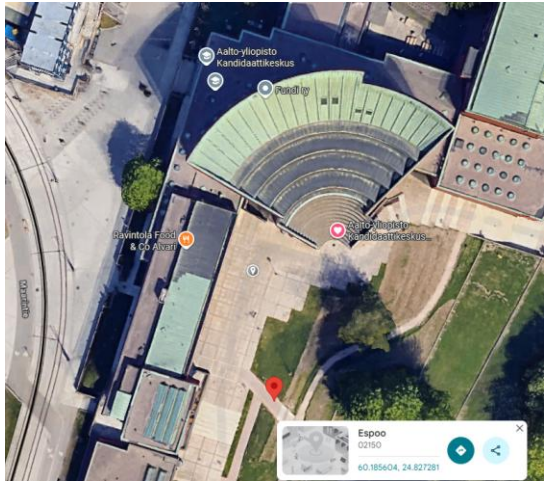
Position:

Latitude: 60.185358188529044

Longitude: 24.82737019658089

Height: 10

GPS from Maps



Quaternion:

X: 0

Y: 0

Z: 1

W: 0

Euler angles (degrees):

East: 0

North: 0

Up: -180

Orientation (ENU)

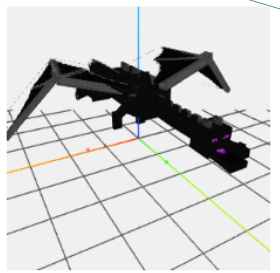
https://raw.githubusercontent.com/balazs-ladjanszki/3d/main/dragon_scaled.glb

Refs :

Content Type: model/gltf+json

URL: https://raw.githubusercontent

URL



model/gltf+json

Add at least one key-value pair
to pass the SCR verification (bug)

Definitions :

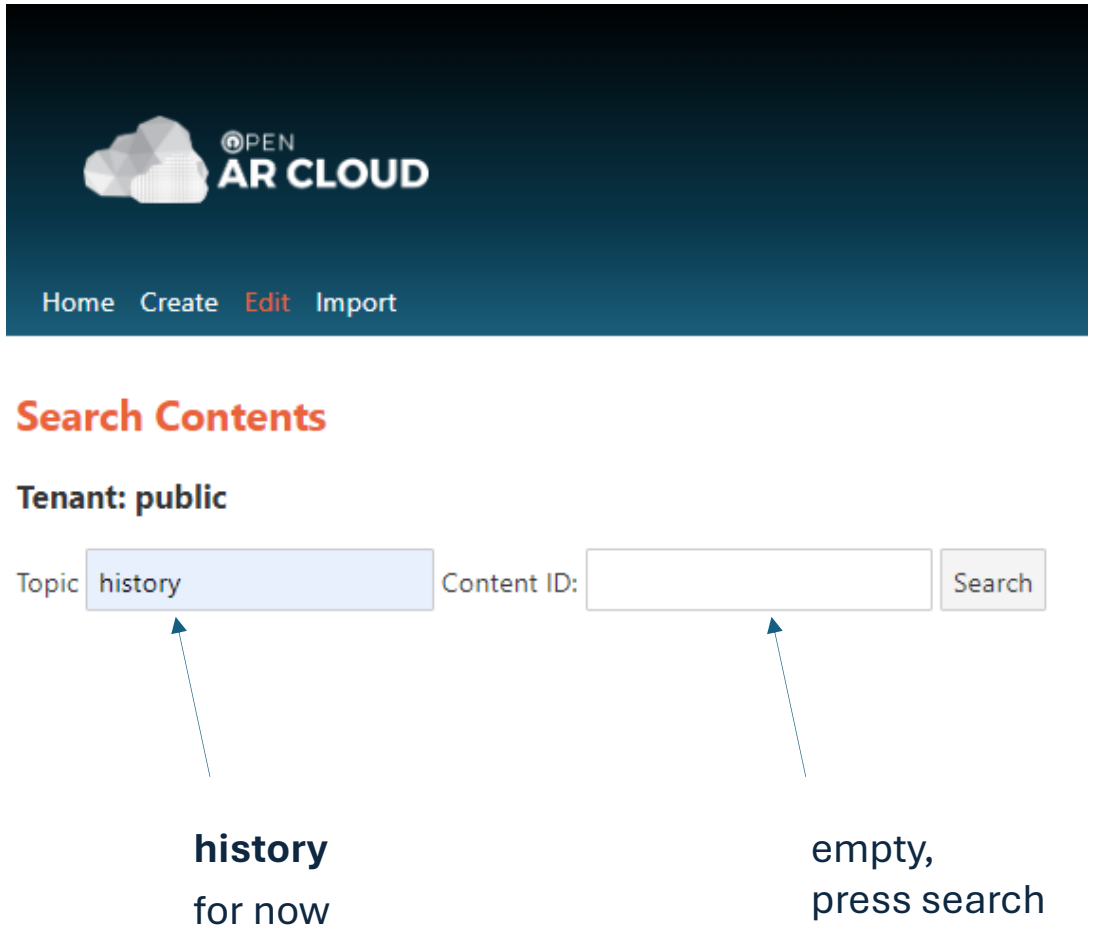
Type: testkey

Value: testvalue

Test content at GMU

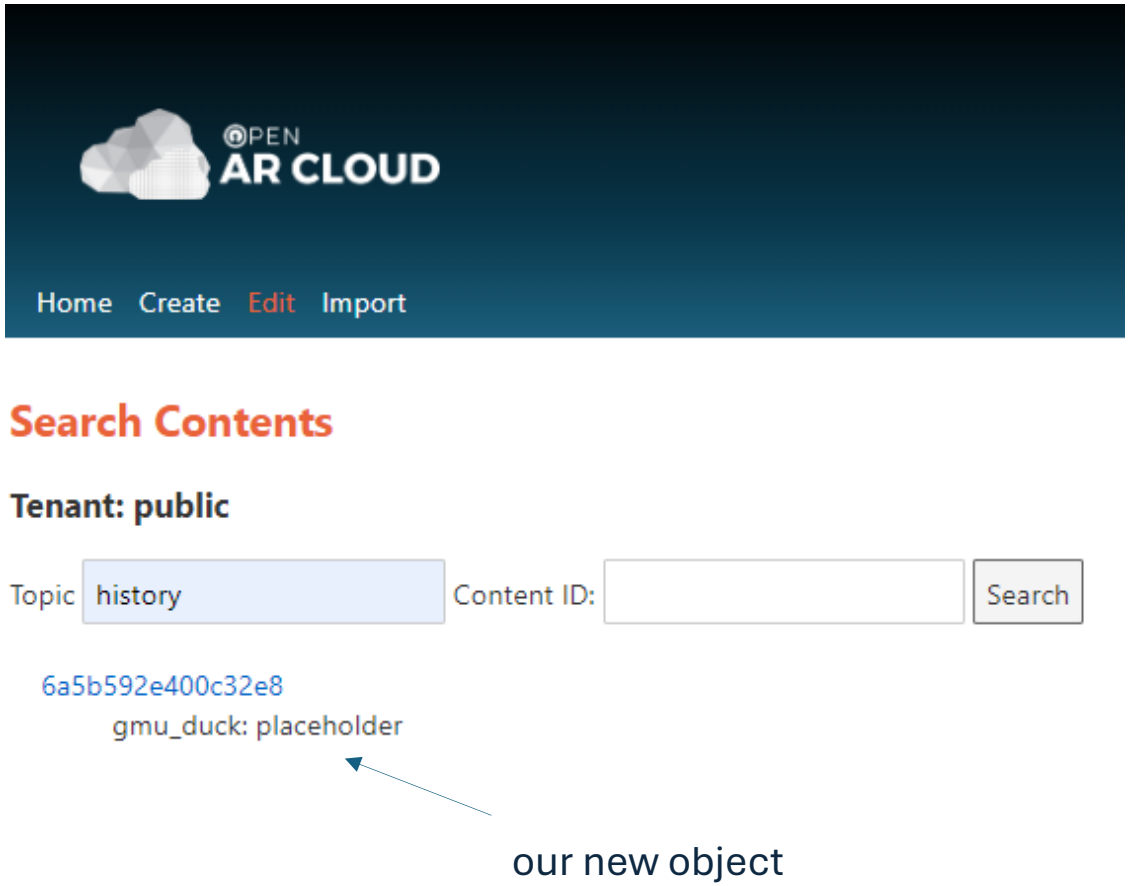
Search and validate

Direct query: https://scd.orbit-lab.org/scrs/history?h3Index=8808996983fffff
note that client apps assemble this URL and use SCD this way



history
for now

empty,
press search



our new object

OGL content

Currently supported contents:

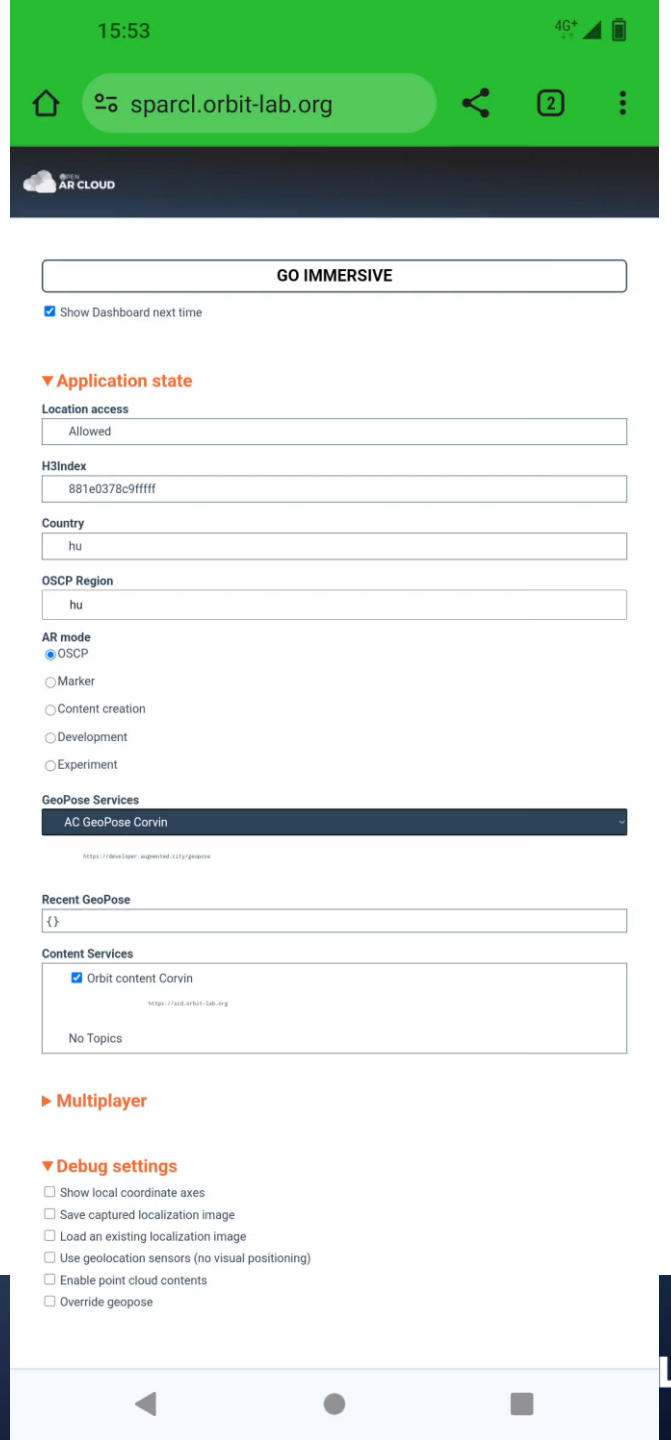
- placeholders
- primitives from textual descriptions
- GLTF/GLB models (texture OK, but no complex shaders)
- iframe content loaded via URL



`https://github.com/oframe/ogl`

OGC Point of Interest

```
{
   "type": "Feature",
   "geometry": {"type": "Point",
   "coordinates": {48.22459095, 16.35021388}},
   "properties": {
      "featureID": 693842,
      "name": {"name" : "Volksoper Wien"},
      "hasFeatureOfInterest": {"href": "https://volksoper.at"},
      "hasPayload": {
        "usesSchema" : "egpoi.json",
        "hasDefinition" : "egpoi",
        "isPublic:" : true
      },
      "openingHours": {
          "openingHoursLines": "Tu-Fr 9:00-17:00",
          "openingHoursFormat": "schemadotorg"
      },
      "category": {
          "category": "theatre",
          "categorySystem": "osm"
      }
   }
}
```



VIDEO: https://youtu.be/7Ua7Tpan2HI

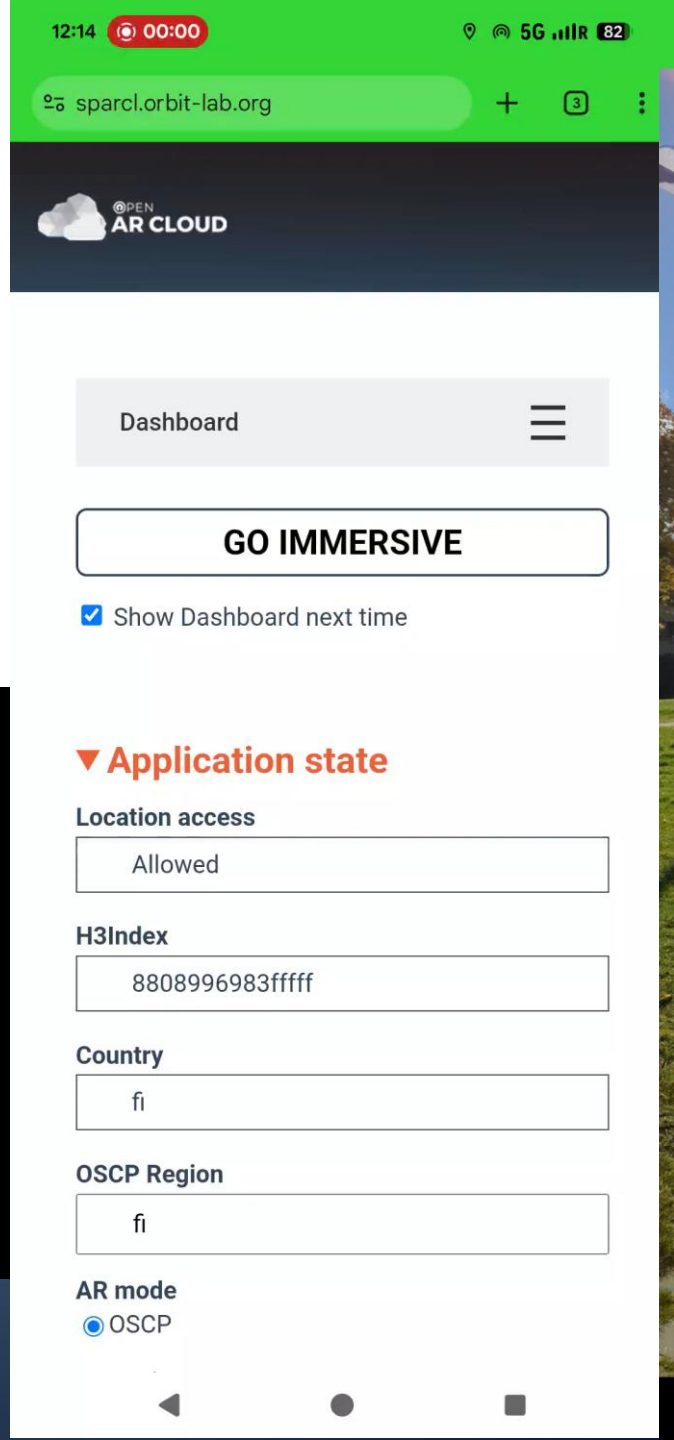
Demo at the campus

https://sparcl.orbit-lab.org

Try it!

Video: https://youtube.com/shorts/P_xfyFADVQU







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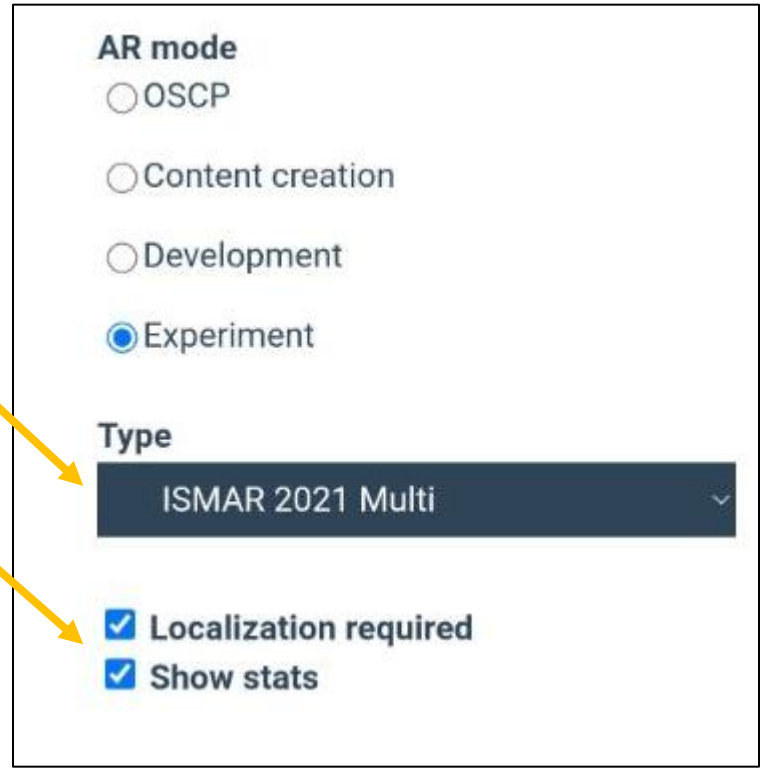
Extending sparcl

Experiment mode:

- define an experiment name and add to `Selector.svelte`
- `Settings.svelte`: this will be loaded into the main Dashboard
- `Viewer.svelte`: extend standard functionality

All our demos are implemented as experiments

(see `ismar2021` branch)



`https://github.com/OpenArCloud/sparcl`

`https://openarcloud.github.io/sparcl/architecture/experiments.html`

context-aware animated content (ThreeJS)



Video: https://youtu.be/23kC43Xg2q4

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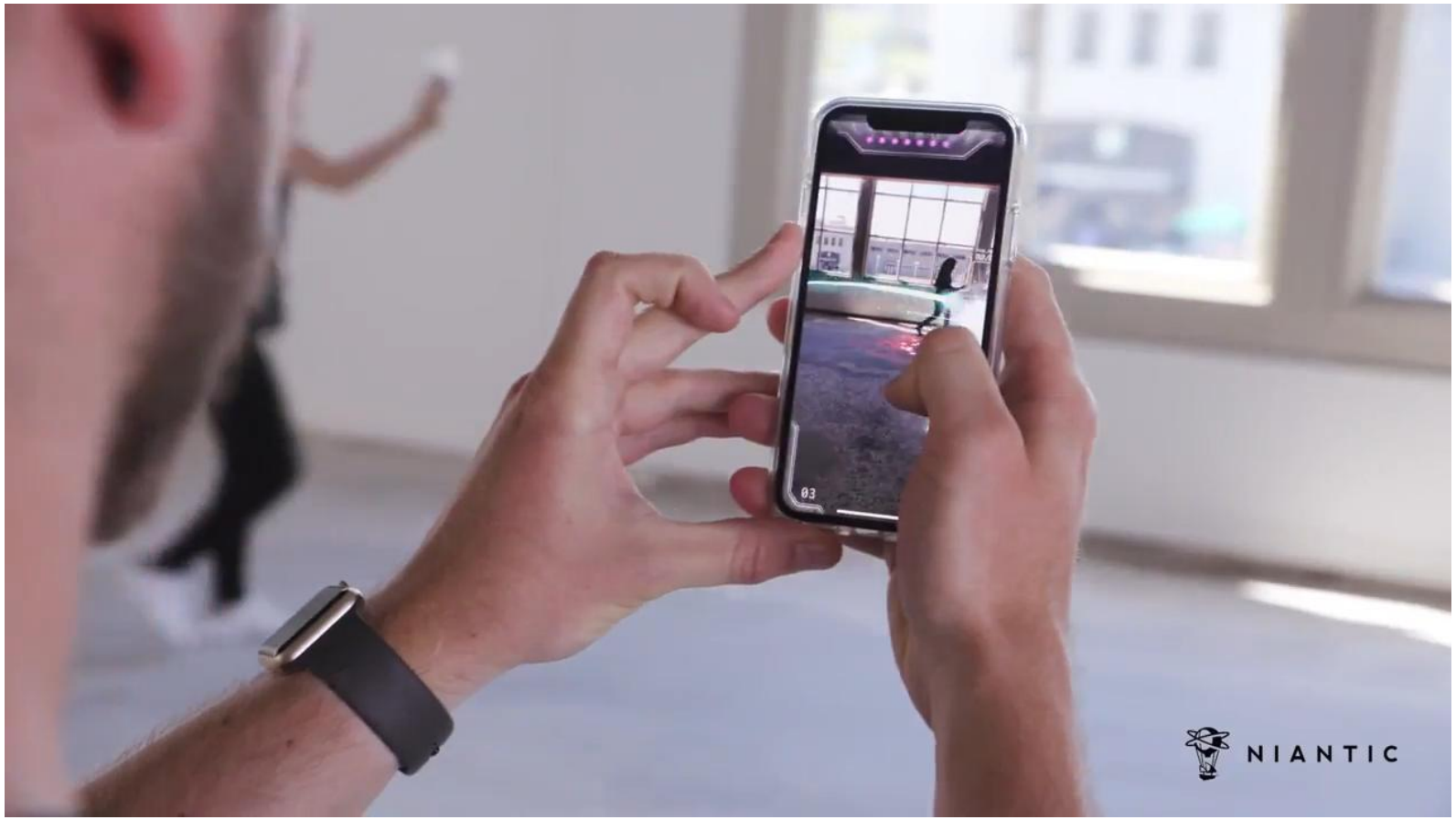
spARcl advanced features

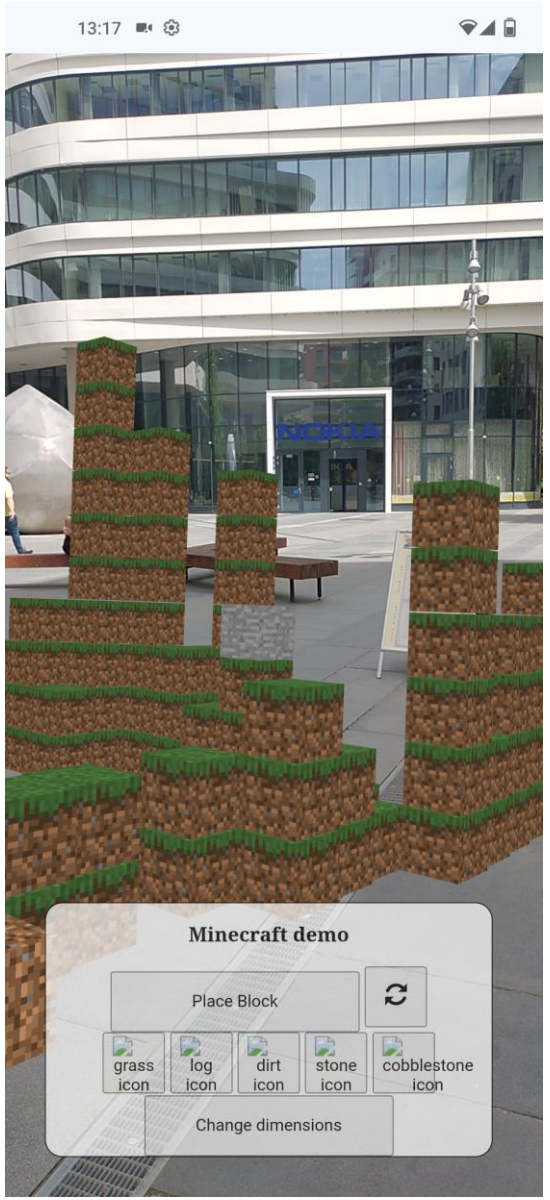
Dr. Gábor Sörös

Nokia Bell Labs & Open AR Cloud

OPEN AR CLOUD

Multiplayer games in augmented reality





Peer-to-peer multiuser experiences

all clients run the same code
multiple transmission options (PeerJS, RabbitMQ, etc.)
local network parameters (e.g., rendezvous server) are retrieved from SSD

events are shared and logged: player poses, objects created, etc.

local-first design (~ collaborative document editing) using Automerge
headless client (lives forever) stores history
- offline periods are OK
- CRDT: conflict-free replicated datatypes
- automatic merging of diverged histories
- every new client can get the full history

https://github.com/OpenArCloud/sparcl/blob/main/docs/guides/multiuser.md

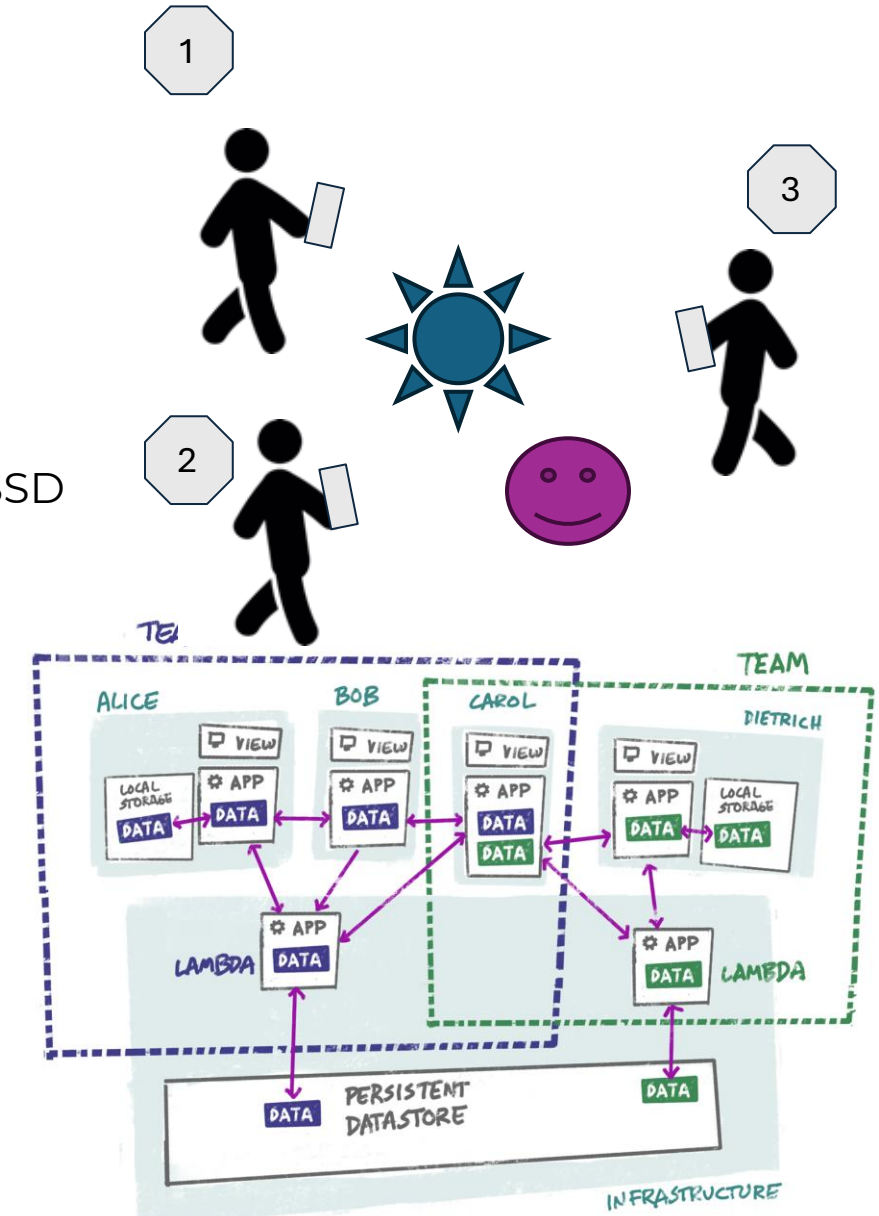
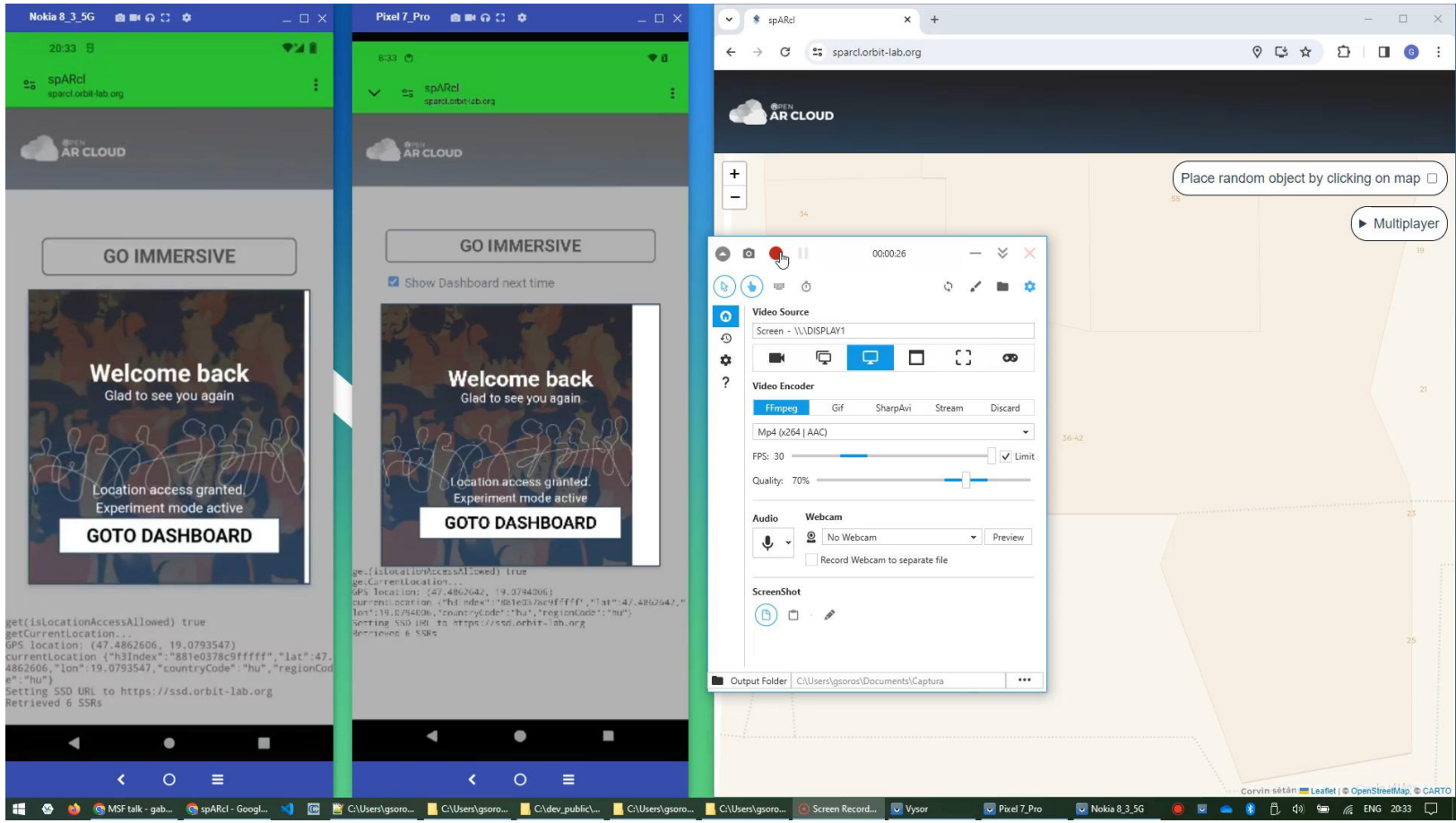


Figure from Herb Caudill, local-first software

Nokia-OARC peer-to-peer AR object sharing

- All devices run own motion tracking, co-localize via VPS into shared geo-coordinate system (GeoPoseProtocol)
- No proprietary AR anchors
  but only GeoPose
- WebXR gives us hit test target point on the ground (local pose). We calculate the GeoPose of the hit point
- We generate a random object descriptor (shape, size, color) as JSON. We fill it into a spatial content record (SCR) with the GeoPose and share it
- We draw the received object by the renderer
- Works in both directions



Video: https://youtu.be/JTqQ0HALyXY?si=Y46ovfkEWIjIVFL0&t=173

Collaborative block-world builder

Code: https://github.com/balazs-ladjanszki/OARC_sparcl/tree/minecraft2

App: https://oarcvps.duckdns.org:8044/

Try it!

Blocks shared over P2P

Players need to start in the same H3 cell

Contents are persisted until one player is online

To restart building, everyone must clear the automerge history





Videos:
https://youtube.com/shorts/RgWFm2iHCRc
https://youtu.be/ixnEXkaeD6c

P2P service and the headless client

We need to register a service of type **`p2p-master`** in the SSD for the coverage area. The clients will receive all necessary information (e.g., signaling server URL and peer id of the headless client to connect to) from the spatial service record.

To keep the history of the experience at a location, a long-running, headless client can be started which merely serves as a bookkeeper. The code for this is already withing sparcl, it is seen as an equal peer in the P2P network but it has no AR features and no UI at all.

Start a headless client:
`https://sparcl.orbit-lab.org?headless=true`
(simply specify **`peerid`** as a parameter)

Start a headless client and use specific signaling server and port:
`https://sparcl.orbit-lab.org?headless=true&signal=` YOUR_URL `u&port=` YOUR_PORT
(besides **`peerid`**, also specify **`signal`** and **`port`**)

GeoPose streams

Try it!

Video: https://youtu.be/Z09ilsnFpSw

Register **Message-Broker** service (RabbitMQ)

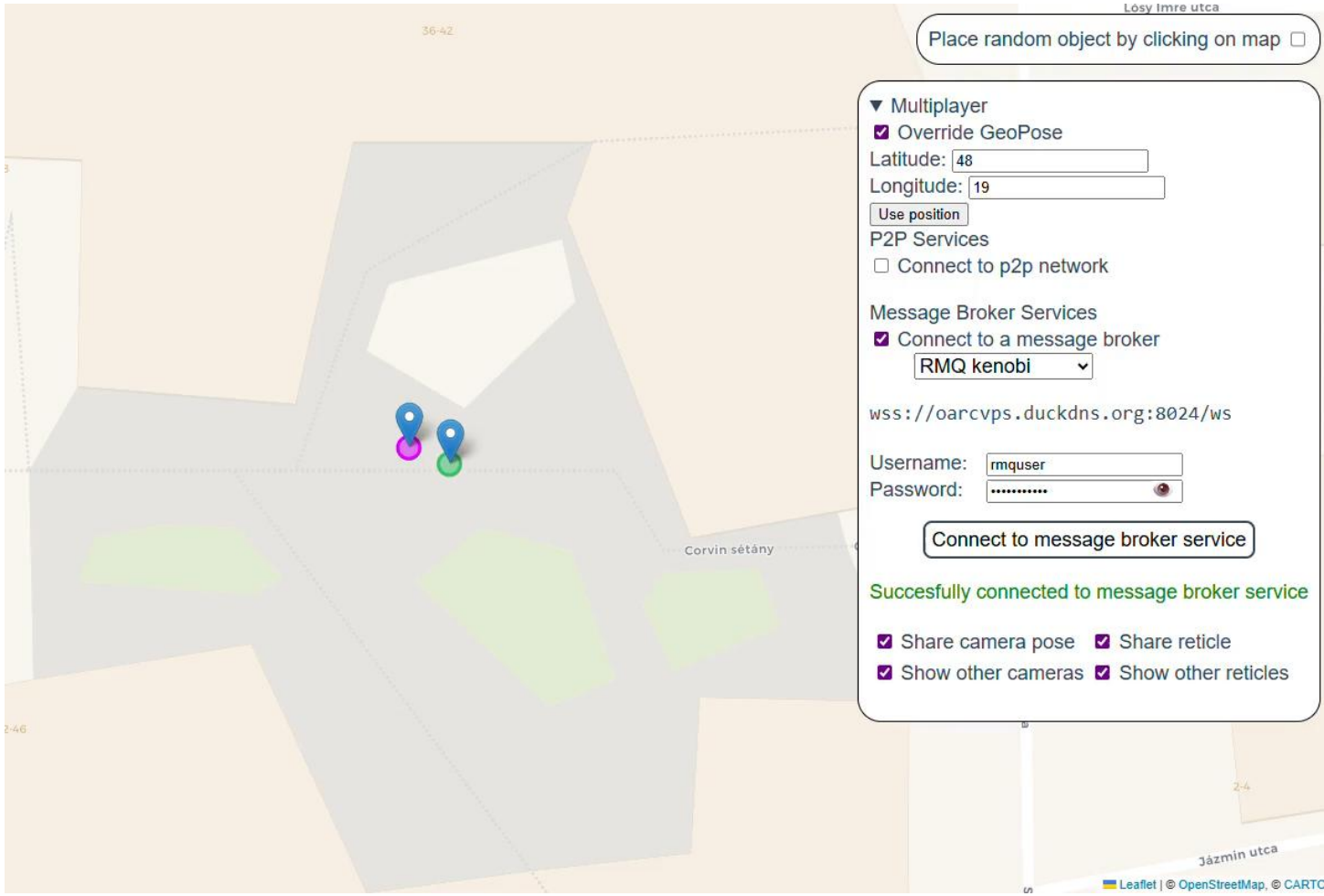
Password protection:
add authentication: password in SSR
(ask me for password)

If service available, spARcl shows RMQ widget

Multiplayer username can be set or extracted from the login email

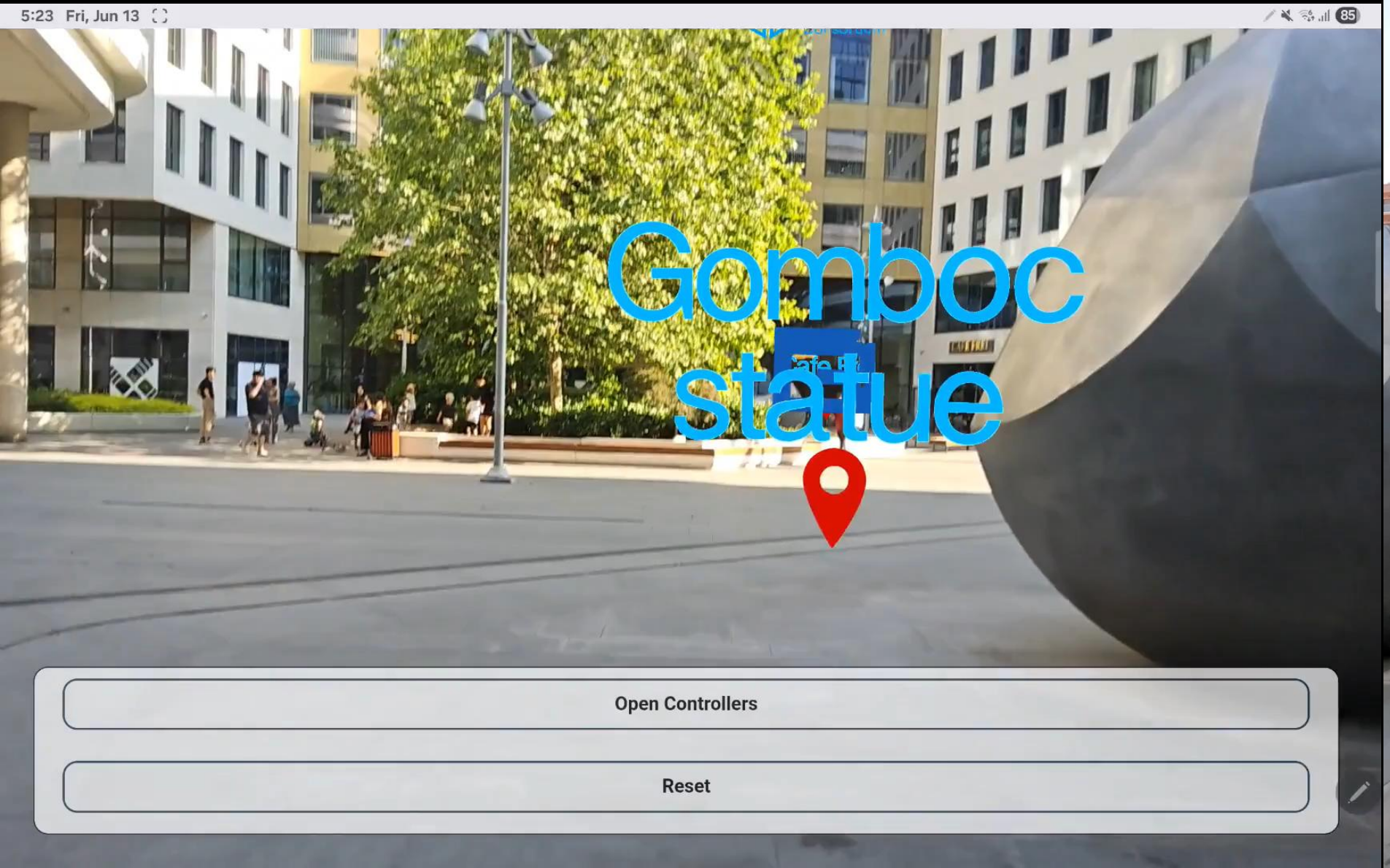
RabbitMQ (AMQP) message broker connection

- Streaming camera GeoPose to others
- Subscribing to GeoPose stream of others, visualizing them in AR (floating name tag, avatar color, etc)
- Other streams: IoT sensor data, robot pose, robot waypoints, etc.



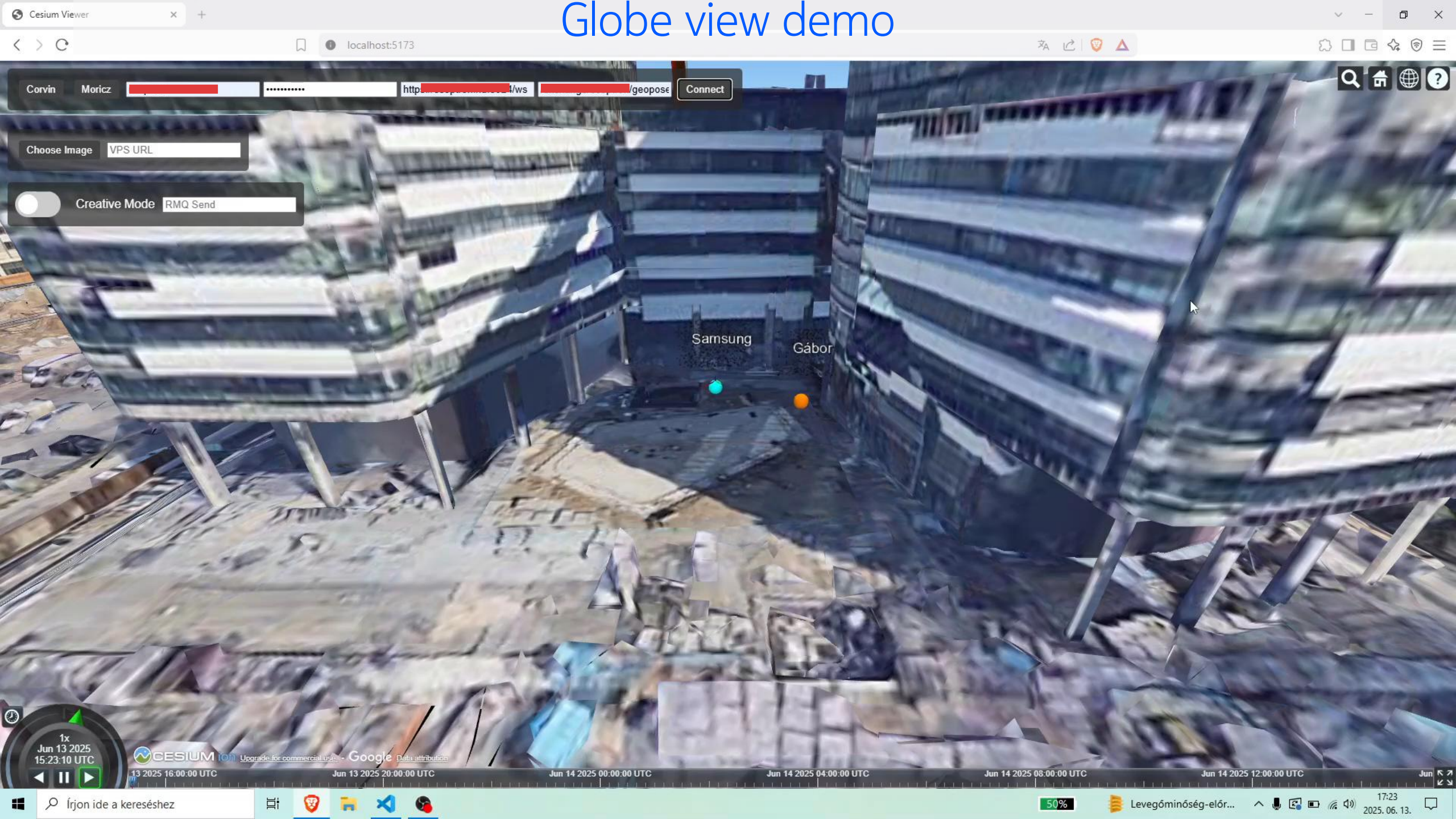
Multiplayer demo

https://youtu.be/0Sjo_IBs1K8
https://youtube.com/shorts/IJRx_ehYzHI





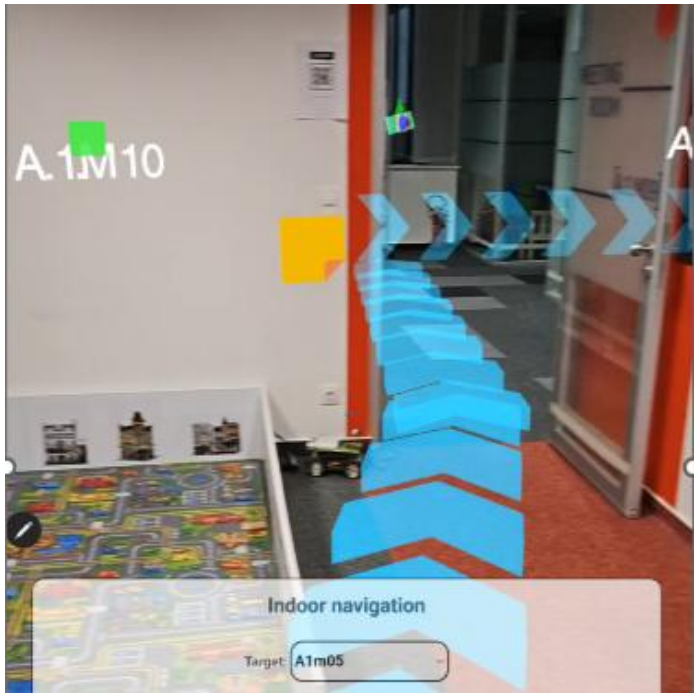
Globe view demo

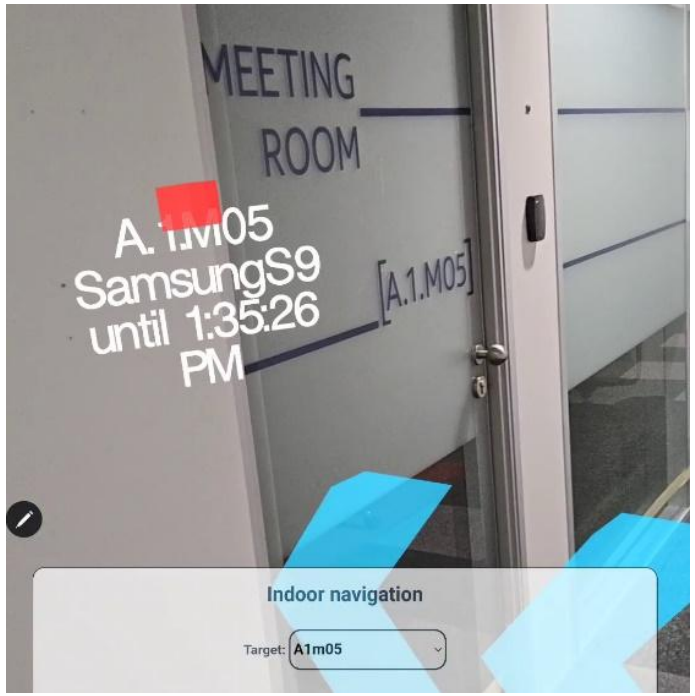


VIDEO: Connecting mobile robots to the open AR cloud
https://youtu.be/KoYgbabxeVs

Nokia Office AR

Indoor localization, Navigation, Seat reservation, Room reservation







Location-anchored spatial sounds

Today, 3D soundscapes can be rendered easily on a smartphone with earplugs

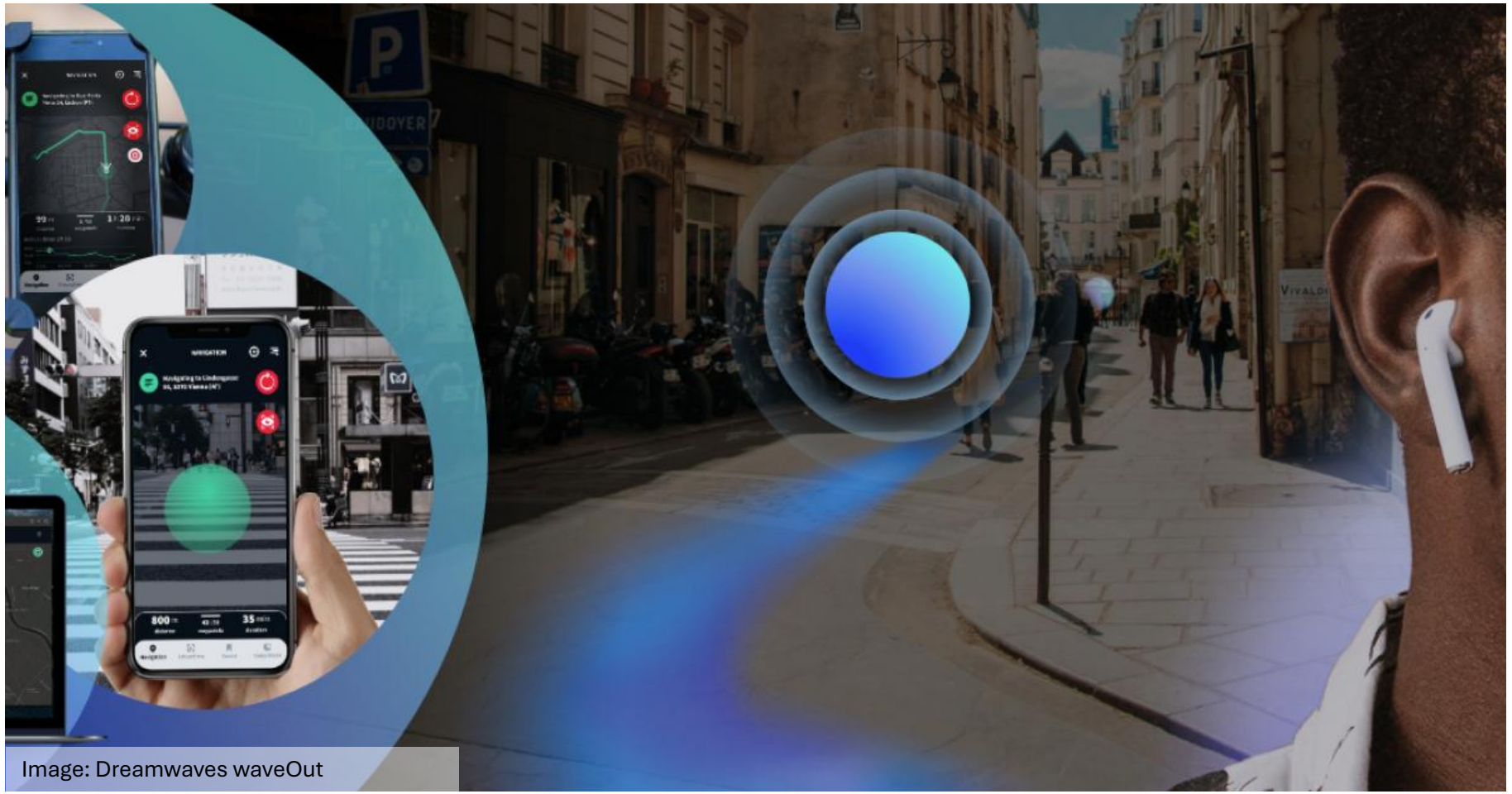
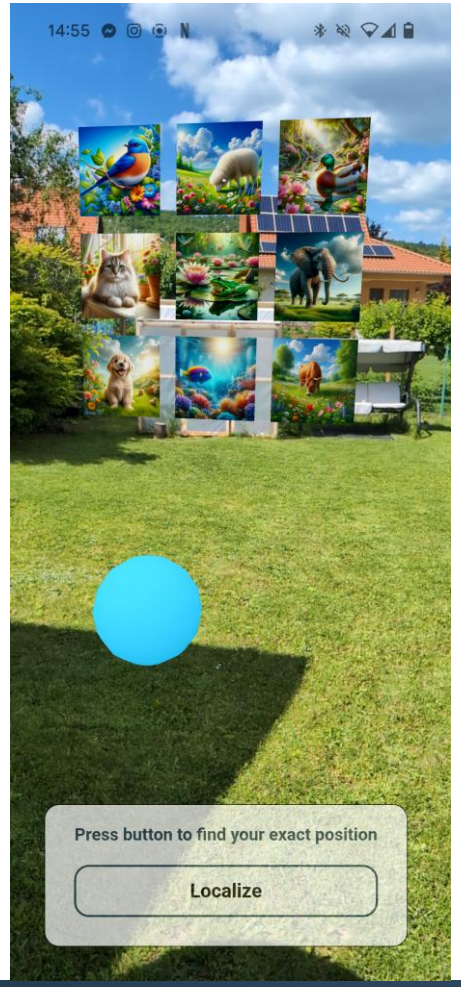


Image: Dreamwaves waveOut

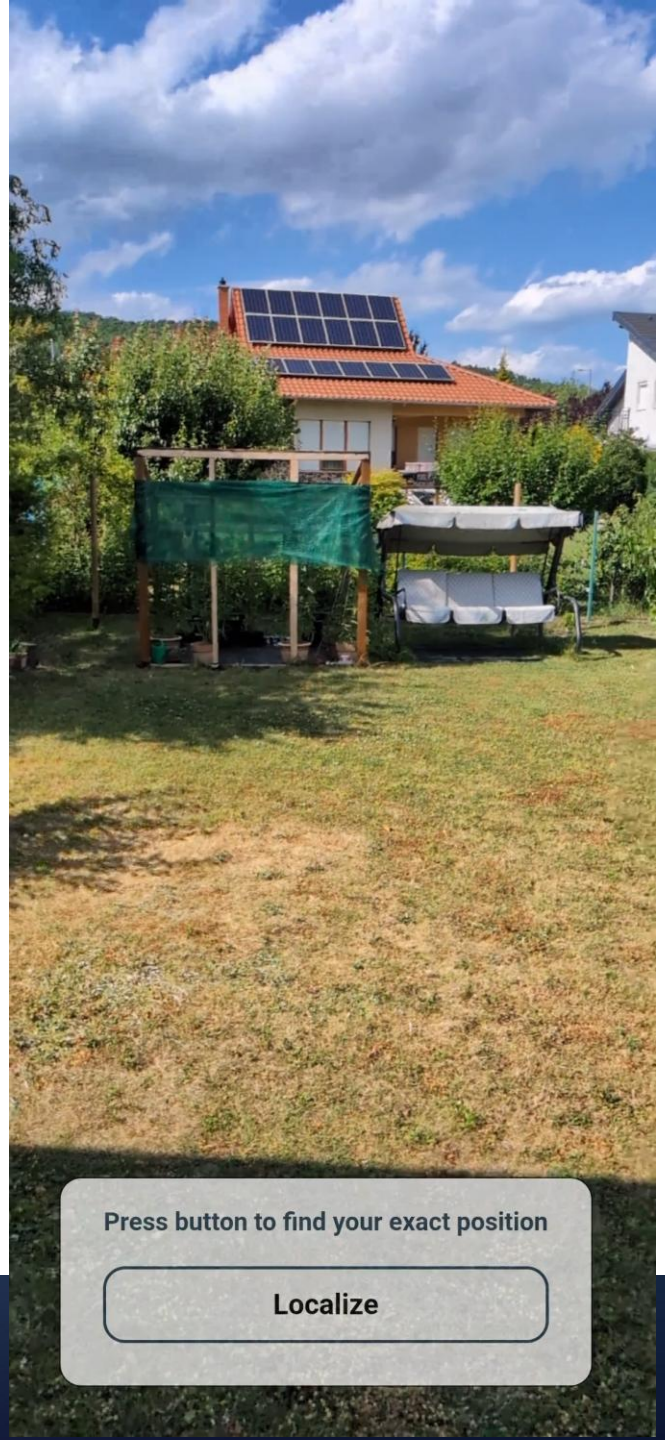
Sound search game

Which animal is in which bush?









Lab project of Imola Szin, TU Budapest, 2025

Contributors:



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Andor Kovács
(engineering intern)

Q & A

Test it at
https://sparcl.orbit-lab.org
spARcl docs: https://openarcloud.github.io/sparcl/
spARcl code: https://github.com/OpenArCloud/sparcl

References

Gábor Sörös, Andor Kovács, Dénes Bisztray, Asmaa Thabet, Balázs Ladjánszki, James Jackson, Christine Perey, Alina Kadlubsky - Demo: Open Visual Positioning and Discovery Services for Location-based Augmented Reality, In Companion of the the 2025 ACM International Joint Conference on Pervasive and Ubiquitous Computing (UbiComp Companion '25), Helsinki, Finland, Oct 2025

Gábor Sörös, James Jackson, Michael Vogt, Mikel Salazar, Alina Kadlubsky, Jan-Erik Vinje - An Open Spatial Computing Platform, In Proc. 2024 IEEE International Conference on Metaverse Computing, Networking, and Applications (MetaCom), Hong Kong, Aug 2024

Gábor Sörös, Sebastian Eger - Poster: Mobile Robots and Sensor Streams in an Open Spatial Computing Platform, In 2024 EuCNC \& 6G Summit, Antwerp, Belgium, June 2024

Gábor Sörös, John Nilsson, Nan Wu, Jennifer Shane, Alina Kadlubsky - Demo: End-to-end open-source location-based augmented reality in 5G, In IEEE Symposium on Mixed and Augmented Reality (ISMAR), Singapore, Oct 2022

James Jackson, Michael Vogt, Gábor Sörös, Mikel Salazar, Sergey Fedorenko - Demo: The first open AR cloud testbed, In IEEE Symposium on Mixed and Augmented Reality (ISMAR), Bari, Italy, Oct 2021

Links

GeoPose 1.0 Website

OGC draft POI Conceptual Model

OARC Demonstration of PoI, VPS and GeoPose

The First Open AR Cloud Testbed (ISMAR 2021)
https://www.youtube.com/watch?v=E-kSY5JaIVA

End-to-End Open-Source Location-based Augmented Reality on 5G (ISMAR 2022)
https://www.youtube.com/watch?v=8BPHI5WMmYI

A WebXR Client for an Open AR Cloud (MSF tutorial) https://youtu.be/wok0clWDtfo

OARC PoI interoperabiltiy demo 2024 https://medium.com/openarcloud/historical-interoperability-demo-for-poi-data-on-oscp-at-ogcs-128th-member-meeting-in-delft-9ca063299d36

Martin Kleppmann - Local-first software: You own your data, in spite of the cloud,
https://www.inkandswitch.com/local-first/

UBICOMP ISWC2025

14-16 Oct. 2025
Helsinki, Finland

OARC Unity client

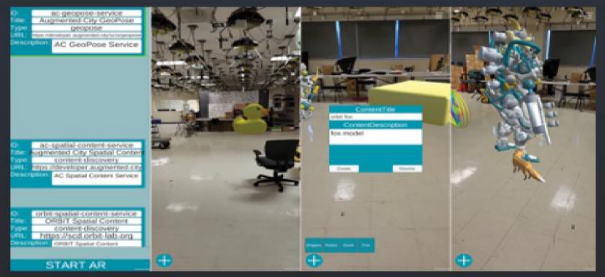
OPEN AR CLOUD

End-to-end open-source location-based augmented reality in 5G

Gábor Sörös, John Nilsson, Nan Wu, Jennifer Shane, Alina Kadlubsky

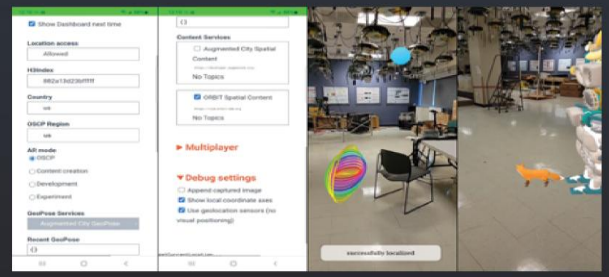
We present a fully open source, end-to-end solution for placing, storing, and re-discovering geo-anchored AR contents in the real world. We advanced components of the Open Spatial Computing Platform (OSCP) and deployed it in a 5G testbed.
We created a client in Unity and built representative demo applications that allow virtual objects to persist throughout space and time. We created a data collector application for spatial mapping, and we connected the OSCP with an open-source visual positioning system. Furthermore, we created tutorials on the setup to foster its adoption in the scientific community as a platform for location-based AR experiments.

Experiment 1



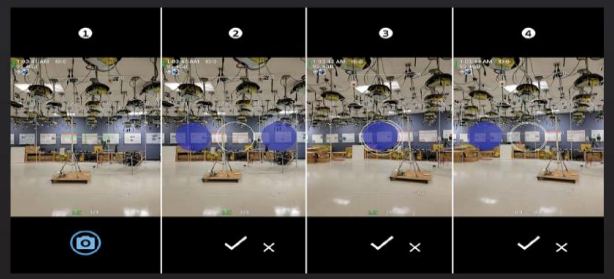
Open-source Unity client (Aurora viewer) localizing in the WINLAB, discovering services, discovering the existing contents, and creating new content (Fox). All objects are linked to GPS coordinates via OGC GeoPose.

Experiment 2



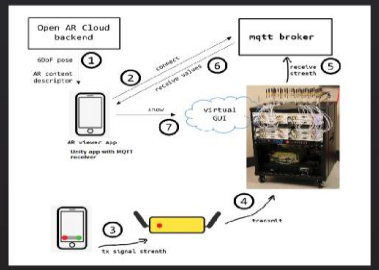
Open-source WebXR client (Sparcl) localizing in the WINLAB, discovering the available services, discovering the existing contents, and discovering the newly created content.

Image capture app for VPS mapping



Open-source camera app showing the compass direction, level lines, and hints about the next best view to capture. When the user rotates the device to align over the next blue spot, the next image is automatically captured. The images (along with GPS coordinates and other metadata) are stored in the universal Kapture format.

Experiment 3



Demonstrating augmented reality visualization of real-time IoT sensor data. We retrieve sensor values, and show them in dynamically generated UI widgets above the physical object. For example the signal strength of a remote-controlled radio.

https://github.com/OpenArCloud